CFA Level I Mock Exam 6.1 - Mark Scheme

Each question lists the correct choice and the original feedback for all three choices. Source excerpts accompany explanations where PDF text extraction may lose equation layout.

Question 1

Max Ohn, CFA, works as a client advisor for a small firm. He has established a referral program with a real estate agency. Ohn refers clients to the real estate agency and the real estate agency refers clients to Ohn. Because the real estate agency does not compete with Ohn's employer and because the arrangement benefits Ohn's firm, he omits disclosure of the arrangement to his firm. Ohn diligently discloses the referral arrangement to all referred clients after they become firm clients. Has Ohn most likely violated the Standard relating to referral fees?

Correct answer: C

A. No - Incorrect

Incorrect Answer Feedback: Incorrect because Standard VI (C), Referral Fees, states that members and candidates must disclose to their employer, clients, and prospective clients, as appropriate, any compensation, consideration, or benefit received from or paid to others for the recommendation of products or services. Members and candidates must disclose when they pay a fee or provide compensation to others who have referred prospective clients to the member or candidate. Therefore, Ohn is required to disclose the arrangement to his employer. Further, he is required to disclose the arrangement before, not after, a referral becomes a client.

B. Yes, only by omitting disclosure of the arrangement to his firm - Incorrect

Incorrect Answer Feedback: Incorrect because Standard VI (C), Referral Fees, states that members and candidates must disclose to their employer, clients, and prospective clients, as appropriate, any compensation, consideration, or benefit received from or paid to others for the recommendation of products or services. Members and candidates must disclose when they pay a fee or provide compensation to others who have referred prospective clients to the member or candidate. Therefore, Ohn is required to disclose the arrangement to his employer. Further, he is required to disclose the arrangement before, not after, a referral becomes a client.

C. Yes, by omitting disclosure of the arrangement to his firm and by disclosing the referral arrangement to referred clients after they become firm clients - Correct

Correct Answer Feedback: Correct because Standard VI (C), Referral Fees, states that members and candidates must disclose to their employer, clients, and prospective clients, as appropriate, any compensation, consideration, or benefit received from or paid to others for the recommendation of products or services. Members and candidates must disclose when they pay a fee or provide compensation to others who have referred prospective clients to the member or candidate. Therefore, Ohn is required to disclose the arrangement to his employer. Further, he is required to disclose the arrangement before, not after, a referral becomes a client: Appropriate disclosure means that members and candidates must advise the client or prospective client, before entry into any formal agreement for services, of any benefit given or received for the recommendation of any services provided by the member or candidate.

Question 2

A member, who is an independent analyst, is approached by two small companies to write issuer-paid research report. Company 1 proposes to pay a flat fee plus stock warrants to the member. Company 2 requires the member to obtain approval prior to publishing the reports. The member most likely violates the Standards if the member agrees to write reports for:

Correct answer: C

A. Company 1 only. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard I(B) Independence and Objectivity, with regards to issuer-paid research, best practice is for independent analysts, prior to writing their reports, to negotiate only a flat fee for their work that is not linked to their conclusions or recommendations. Therefore, accepting the compensation from Company 1 will result in a violation. Furthermore, accepting the compensation from Company 2 will also result in a violation. According to Standard I(B) Independence and Objectivity, Members and Candidates

must use reasonable care and judgment to achieve and maintain independence and objectivity in their professional activities. Prior approval of the research reports by the covered company before they are published will impact the member's objectivity.

B. Company 2 only. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard I(B) Independence and Objectivity, Members and Candidates must use reasonable care and judgment to achieve and maintain independence and objectivity in their professional activities. Prior approval of the research reports by the covered company before they are published will impact the member's objectivity. Therefore, accepting the compensation from Company 2 will result in a violation. Furthermore, accepting the compensation from Company 1 will also result in a violation. According to Standard I(B) Independence and Objectivity, with regards to issuer-paid research, best practice is for independent analysts, prior to writing their reports, to negotiate only a flat fee for their work that is not linked to their conclusions or recommendations.

C. both Company 1 and Company 2. - Correct

Correct Answer Feedback: Correct because according to Standard I(B) Independence and Objectivity, Members and Candidates must use reasonable care and judgment to achieve and maintain independence and objectivity in their professional activities. Members and Candidates must not offer, solicit, or accept any gift, benefit, compensation, or consideration that reasonably could be expected to compromise their own or another's independence and objectivity. With regards to issuer-paid research, best practice is for independent analysts, prior to writing their reports, to negotiate only a flat fee for their work that is not linked to their conclusions or recommendations. Furthermore, prior approval of the research reports by the covered company before they are published will impact the member's objectivity.

Question 3

Toni Lynn, CFA, is a research analyst. Lynn relies on her colleagues to determine whether the third-party research she uses is sound. Lynn also uses content from personal blogs and social media websites for her research after concluding it is sound. Has Lynn most likely violated the Standard relating to diligence and reasonable basis?

Correct answer: A

A. No - Correct

Correct Answer Feedback: Correct because according to Standard V(A), Diligence and Reasonable Basis, a member or candidate may rely on others in his or her firm to determine whether secondary or third-party research is sound and use the information in good faith unless the member or candidate has reason to question its validity or the processes and procedures used by those responsible for the research. Also, according to Standard V(A), Diligence and Reasonable basis, the sources of the information and data will influence the level of the review a member or candidate must undertake. Information and data taken from internet sources, such as personal blogs, independent research aggregation websites, or social media websites, likely require a greater level of review than information from more established research organizations. Because relying on her colleagues to check third-party research is permitted; and Standard V(A) does not prohibit using personal blogs and social media websites for her research, Lynn has not violated Standard V(A).

B. Yes, by using content from personal blogs and social media websites for her research - Incorrect

Incorrect Answer Feedback: Incorrect because Lynn most likely has not violated Standard V(A), Diligence and Reasonable Basis, as described in the response rationale of the correct answer.

C. Yes, by relying on her colleagues to determine whether the third-party research is sound - Incorrect

Incorrect Answer Feedback: Incorrect because Lynn most likely has not violated Standard V(A), Diligence and Reasonable Basis, as described in the response rationale of the correct answer.

Question 4

Teresa Avila, CFA, is a micro cap investment analyst at a hedge fund. The fund requires Avila to hold any securities she recommends for the fund in her own account as well. Because Avila has such a small account, whenever she trades for her own portfolio she combines the transactions with those of the hedge fund so she

is sure to have her account aligned with the fund. Has Avila most likely violated any CFA Institute Standards of Professional Conduct?

Correct answer: C

A. No - Incorrect

Incorrect Answer Feedback: Incorrect because Avila has violated Standard VI(B), Priority of Transactions.

B. Yes, the Standard relating to misconduct - Incorrect

Incorrect Answer Feedback: Incorrect because Avila has not violated Standard I(D), Misconduct.

C. Yes, the Standard relating to priority of transactions - Correct

Correct Answer Feedback: Correct because Standard VI(B), Priority of Transactions, requires that investment transactions for clients and employers have priority over transactions in which members have beneficial ownership. By executing her own accounts transactions with those of the hedge fund the analyst has violated this Standard as micro cap securities can be thinly traded and easily influenced by changes in the volume of activity. So the analyst may benefit when she combines her transactions with the hedge funds and she should let the fund execute its orders before she makes changes to her account.

Question 5

Kush Shah, CFA, is the chief investment officer of an investment firm. Shah discovers during a routine review of the firm's trading system that the firm erroneously bought shares of IY Steel (IYS) for several clients. Shah immediately sells IY shares and realises significant profits for the clients. Later, Shah outsources a portion of the clients' portfolios to an external manager. Shah does not notify the firm's clients about the external manager because the fund's stock selection process remains unchanged. Shah has violated the Standards because he failed to notify the clients:

Correct answer: C

A. only about the erroneous purchase of IY shares. - Incorrect

Incorrect Answer Feedback: Incorrect because Shah violated the Standard by failing to notify clients both about the erroneous purchase of IY shares and about outsourcing a portion of the clients' portfolios to an external manager, as described in the response rationale to the correct answer.

B. only about outsourcing a portion of the clients' portfolios to an external manager. - Incorrect

Incorrect Answer Feedback: Incorrect because Shah violated the Standard by failing to notify clients both about the erroneous purchase of IY shares and about outsourcing a portion of the clients' portfolios to an external manager, as described in the response rationale to the correct answer.

C. about the erroneous purchase of IY shares and about outsourcing a portion of the clients' portfolios to an external manager. - Correct

Correct Answer Feedback: Correct because according to Standard V(B) relating to Communication with Clients and Prospective Clients, the member or candidate must keep clients and other interested parties informed on an ongoing basis about changes to the investment processes, specially newly identified significant risks and limitations. Thus, the outsourcing to the external manager should be notified to clients. Also Shah, violated V(B) by not disclosing the error in the trading system. Clients should have been informed about the error and the corrective actions the firm was undertaking on their behalf. Hence, both actions constitute a violation of the Standards.

Question 6

The Standard relating to additional compensation arrangements requires members to disclose which of the following that might create a conflict with their employer's interests?

Correct answer: C

A. Only direct compensation - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard IV (B), Additional Compensation Arrangements, members must obtain permission from their employer before accepting compensation or other benefits from third parties for the services rendered to the employer or for any services that might create a conflict with their employer's interest. Compensation and benefits include direct compensation by the client and any indirect compensation or other benefits received from third parties.

B. Only indirect compensation - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard IV (B), Additional Compensation Arrangements, members must obtain permission from their employer before accepting compensation or other benefits from third parties for the services rendered to the employer or for any services that might create a conflict with their employer's interest. Compensation and benefits include direct compensation by the client and any indirect compensation or other benefits received from third parties.

C. Both direct compensation and indirect compensation - Correct

Correct Answer Feedback: Correct because according to Standard IV (B), Additional Compensation Arrangements, members must obtain permission from their employer before accepting compensation or other benefits from third parties for the services rendered to the employer or for any services that might create a conflict with their employer's interest. Compensation and benefits include direct compensation by the client and any indirect compensation or other benefits received from third parties.

Question 7

Ana Kloes, CFA, passes all three levels of the CFA exam in three consecutive years. After passing the Level III exam, she adds to her email signature, "Partial CFA, awaiting CFA charter." Before Kloes is awarded the CFA charter, she writes on her social media page, "I passed all three levels of the CFA Program on my first attempt in three consecutive years." Has Kloes violated the Standards?

Correct answer: B

A. No - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VII (B), Reference to CFA Institute, the CFA Designation, and the CFA Program, candidates must never state or imply that they have a partial designation as a result of passing one or more levels or cite an expected completion date of any level of the CFA Program. Final award of the charter is subject to meeting the CFA Program requirements and approval by the CFA Institute Board of Governors. Therefore she has violated the Standard by stating partial designation.

B. Yes, because she adds to her email signature, "Partial CFA, awaiting CFA charter" - Correct

Correct Answer Feedback: Correct because according to Standard VII (B), Reference to CFA Institute, the CFA Designation, and the CFA Program, candidates must never state or imply that they have a partial designation as a result of passing one or more levels or cite an expected completion date of any level of the CFA Program. Final award of the charter is subject to meeting the CFA Program requirements and approval by the CFA Institute Board of Governors. Therefore she has violated the Standard by stating partial designation.

C. Yes, because she adds on her social media page, "I passed all three levels of the CFA Program on my first attempt in three consecutive years" - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VII (B), Reference to CFA Institute, CFA Designation, and CFA Program, this is not a violation. If a candidate passes each level of the exam in consecutive years and wants to state that he or she did so, that is not a violation of Standard VII(B) because it is a statement of fact. If the candidate then goes on to claim or imply superior ability by obtaining the designation in only three years, however, he or she is in violation of Standard VII(B).

Question 8

To be compliant with the GIPS standards, after a firm presents five years of performance history, it must build up to a minimum of:

Correct answer: B

A. 7 years of compliant performance presentation. - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the response rationale for the correct answer, after a firm presents a minimum of five years of GIPS-compliant performances, the firm must present an additional year of performance each year, building up to a minimum of 10 years of GIPS-compliant performance, not 7 years of compliant performance.

B. 10 years of compliant performance presentation. - Correct

Correct Answer Feedback: Correct because the GIPS standards state: A firm is required to initially present, at a minimum, five years of annual investment performance that is compliant with the GIPS standards. If the composite or pooled fund has been in existence less than five years, the firm must present performance since the composite or pooled fund inception date. Prospectively, the firm must present an additional year of performance each year, building up to a minimum of 10 years of GIPS-compliant performance.

C. 15 years of compliant performance presentation. - Incorrect

Incorrect Answer Feedback: Incorrect because, as described in the response rationale for the correct answer, after a firm presents a minimum of five years of GIPS-compliant performances, the firm must present an additional year of performance each year, building up to a minimum of 10 years of GIPS-compliant performance, not 15 years of compliant performance.

Question 9

Akash Gupta has just finished taking the Level II CFA exam for the second time. To protect the integrity of the exam, Gupta is careful not to discuss the exam questions with other candidates. Gupta calls his friend, a member, and contrasts his two attempts by highlighting that there were many calculation questions on derivatives in the first instance and none in the second. Later, in a public blog on investment education, Gupta shares his strong disagreement regarding CFA Institute shifting to computer-based testing. Has Gupta violated the Standards?

Correct answer: B

A. No - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VII(A) relating to Conduct as Participants in CFA Institute Programs, examples of information that cannot be disclosed by candidates sitting for an exam include but are not limited to broad topical areas and formulas tested or not tested on the exam. Gupta has violated Standard VII(A).

B. Yes, by highlighting that there were no calculation questions on derivatives in the recent exam - Correct

Correct Answer Feedback: Correct because according to Standard VII(A) relating to Conduct as Participants in CFA Institute Programs, examples of information that cannot be disclosed by candidates sitting for an exam include but are not limited to broad topical areas and formulas tested or not tested on the exam. Gupta has violated Standard VII(A).

C. Yes, by expressing his disagreement in a public blog regarding CFA Institute shifting to computer-based testing - Incorrect

Incorrect Answer Feedback: Incorrect because Standard VII(A) relating to Conduct as Participants in CFA Institute Programs does not cover expressing opinions regarding CFA Institute, the CFA Program, or other CFA Institute programs. Members and candidates are free to disagree and express their disagreement with CFA Institute on its policies, its procedures, or any advocacy positions taken by the organization.

Question 10

A member who pays a higher brokerage commission on behalf of all of his clients than he would normally pay, without any corresponding benefit to the client, violates the Standard relating to:

Correct answer: C

A. suitability. - Incorrect

Incorrect Answer Feedback: Incorrect because Standard III (C), Suitability, requires that members and candidates who are in an investment advisory relationship with clients consider carefully the needs, circumstances, and objectives of the clients when determining the appropriateness and suitability of a given investment or course of investment action. This scenario does not involve investment recommendation. The member violated Standard III(A), Loyalty, Prudence, and Care.

B. fair dealing. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III (B), Fair Dealing, Members and Candidates must deal fairly and objectively with all clients when providing investment analysis, making investment recommendations, taking investment action, or engaging in other professional activities. All clients are treated in the same way, so the Standard relating to fair dealing is not violated. The member violated Standard III(A), Loyalty, Prudence and Care.

C. loyalty, prudence, and care. - Correct

Correct Answer Feedback: Correct because according to Standard III(A), Loyalty, Prudence, and Care, a member or candidate who pays a higher brokerage commission than he or she would normally pay to allow for the purchase of goods or services, without corresponding benefit to the client, violates the duty of loyalty to the client.

Question 11

Which of the following statements is accurate?

Correct answer: C

A. Verification eliminates the need for in-depth due diligence of a GIPS- compliant firm for a prospective investor. - Incorrect

Incorrect Answer Feedback: Incorrect because the GIPS standards certainly do not eliminate the need for in-depth due diligence on the part of the investor, but compliance with the Standards enhances the credibility of investment management firms that have chosen to undertake this responsibility. Also, once a firm claims compliance with the standards, they may voluntarily hire an independent third party to perform a verification in order to increase confidence in the firm's claim of compliance. Therefore, verification enhances confidence in a firm's claim of compliance and compliance with the GIPS standards enhances credibility of the firm, but does not eliminate the need for in-depth due diligence of the firm for a prospective investor.

B. Verification can be performed on a specific composite to ensure accuracy of the composite presentation. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the GIPS standards, verification is performed with respect to an entire firm, not on specific composites. Verification does not ensure the accuracy of any specific composite presentation.

C. Verification provides assurance on whether the firm's policies and procedures related to pooled fund maintenance have been implemented on a firm-wide basis. - Correct

Correct Answer Feedback: Correct because according to the GIPS standards, verification provides assurance on whether the firm's policies and procedures related to composite and pooled fund maintenance, as well as the calculation, presentation, and distribution of performance, have been designed in compliance with the GIPS standards and have been implemented on a firm-wide basis.

Question 12

Which of the following is not a recommended procedure for compliance with the Standard relating to misrepresentation? Firms should:

Correct answer: A

A. avoid verifying information from third-party firms for timely trade executions. - Correct

Correct Answer Feedback: Correct because when providing information to clients from a third party, members and candidates share a responsibility for the accuracy of the marketing and distribution materials that pertain to the third party's capabilities, services, and products. Misrepresentation by third parties can damage the member's or candidate's reputation, the reputation of the firm, and the integrity of the capital markets. Members and candidates should encourage their employers to develop procedures for verifying information of third-party firms. So, the recommendation is to verify information, but not to avoid verifying it.

B. specifically designate which employees are authorized to speak on behalf of the firm. - Incorrect

Incorrect Answer Feedback: Incorrect because firms can also help prevent misrepresentation by specifically designating which employees are authorized to speak on behalf of the firm.

C. periodically review employee correspondence that contains representations of individual qualifications. - Incorrect

Incorrect Answer Feedback: Incorrect because firms can assist member and candidate compliance by periodically reviewing employee correspondence and documents that contain representations of individual or firm qualifications.

Question 13

A member who is an investment advisor receives a large unsolicited trade request from a client that is not aligned with the client's investment objectives. Which of the following actions by the member is consistent with the Standard relating to suitability? The member:

Correct answer: C

A. immediately executes the trade. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III (C), Suitability, in cases of unsolicited trade requests that a member or candidate knows are unsuitable for a client, the member or candidate should refrain from making the trade until he or she discusses the concerns with the client. Therefore, the member should not immediately proceed with executing the trade.

B. reduces the size of the requested investment and executes the trade. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III (C), Suitability, in cases of unsolicited trade requests that a member or candidate knows are unsuitable for a client, the member or candidate should refrain from making the trade until he or she discusses the concerns with the client. Therefore, the member should not immediately proceed with executing the trade. Although the curriculum discusses smaller unsolicited trade requests as being less serious than large trades, it does not suggest that members unilaterally reduce the size of the trade and execute it without discussion with the client.

C. discusses the investment with the client, adjusts the client's investment policy statement, and executes the trade. - Correct

Correct Answer Feedback: Correct because according to Standard III (C), Suitability, in discussing the trade, the member or candidate should focus on educating the investor on how the request deviates from the current policy statement. Should the unsolicited request be expected to have a material impact on the portfolio, the member or candidate should use this opportunity to update the investment policy statement. Therefore, this member action is consistent with the Standard.

Question 14

When a client asks her how she makes investment decisions, Petra Vogler, CFA, tells the client she uses mosaic theory. According to Vogler, the theory involves analyzing public and nonmaterial nonpublic information including the evaluation of statements made to her by company insiders in one-on-one meetings where management discusses new earnings projections not known to the public. Vogler also gathers general industry information from industry experts she has contacted. Vogler violates the CFA Institute Standards of Professional Conduct because of her use of:

Correct answer: B

A. industry expert information. - Incorrect

Incorrect Answer Feedback: Incorrect because under the mosaic theory it is acceptable to use information from industry contacts as long as the analyst uses appropriate methods to arrive at her conclusions.

B. one-on-one meeting information. - Correct

Correct Answer Feedback: Correct because Vogler violated Standard II(A), Material Nonpublic Information, by using information not known to the public that is selectively disclosed by company insiders in one-on-one meetings. Earnings estimates would likely be considered material and nonpublic information. Information made available to analysts remains nonpublic until it is made available to investors in general. Under the mosaic theory it is acceptable to use information from industry contacts as long as the analyst uses appropriate methods to arrive at her conclusions.

C. nonmaterial nonpublic information. - Incorrect

Incorrect Answer Feedback: Incorrect because it is acceptable to use nonmaterial nonpublic information in her analysis, and this use is not a violation of Standard II(A), Material Nonpublic Information.

Question 15

Shruti Kurup, CFA, is preparing to leave her firm and join a new employer. Prior to leaving the firm, Kurup calls all her clients informing them of her decision to leave. When asked by the clients, Kurup states lack of confidence in the firm's leadership as the reason for her departure and also mentions that more employees are likely to leave soon. Kurup has violated the Standard(s) relating:

Correct answer: A

A. only to loyalty. - Correct

Correct Answer Feedback: Correct because Standard IV(A): Duties to Employers, Loyalty states that CFA Institute members and candidates must act for the benefit of their employer and not otherwise cause harm to their employer. Kurup violates her duty of loyalty to her employer by making disparaging and harmful statements about the firm to its clients. Kurup can call her clients to inform of her decision but cannot make disparaging statements about her employer. Also according to Standard V(B) relating to communication with clients and prospective clients, members and candidates should communicate in a recommendation the factors that were instrumental in making the investment recommendation. There is no investment recommendation being discussed here and hence the Standard is not violated. Only the Standard relating to loyalty has been violated.

B. only to communication with clients and prospective clients. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard V(B) relating to communication with clients and prospective clients, members and candidates should communicate in a recommendation the factors that were instrumental in making the investment recommendation. There is no investment recommendation being discussed here and hence the Standard is not violated.

C. both to loyalty and to communication with clients and prospective clients only. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard V(B) relating to communication with clients and prospective clients, members and candidates should communicate in a recommendation the factors that were instrumental in making the investment recommendation. There is no investment recommendation being discussed here and hence the Standard is not violated.

Question 16

To comply with the Standard relating to knowledge of the law, members are required to take which of the following actions? Action 1: Acquire detailed knowledge of all the laws relating to their activities Action 2: Dissociate from activities of others that they believe are unethical

Correct answer: B

A. Action 1 only - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard I(A), Knowledge of the Law, members and candidates must understand the applicable laws and regulations of the countries and jurisdictions where they engage in professional activities. However, members and candidates are not required to have detailed knowledge of or be experts on all the laws that could potentially govern their activities. So, members are not required to take Action 1.

B. Action 2 only - Correct

Correct Answer Feedback: Correct because according to Standard I(A), Knowledge of the Law, if a member or candidate has reasonable grounds to believe that imminent or ongoing client or employer activities are illegal or unethical, the member or candidate must dissociate, or separate, from the activity. So, members are required to take Action 2.

C. Both Action 1 and Action 2 - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard I(A), Knowledge of the Law, members and candidates must understand the applicable laws and regulations of the countries and jurisdictions where they engage in professional activities. However, members and candidates are not required to have detailed knowledge of or be experts on all the laws that could potentially govern their activities. So, members are not required to take Action 1. Also, if a member or candidate has reasonable grounds to believe that imminent or ongoing client or employer activities are illegal or unethical, the member or candidate must dissociate, or separate, from the activity. So, members are required to take only Action 2.

Question 17

Peter Chang, CFA, spreads rumors on social media about a potential acquisition of Advanced Electronics Company (AEC) after buying the stock for his personal account. Despite the rumors, AEC's stock price declines and Chang closes his position at a significant loss. Has Chang violated the Standards?

Correct answer: C

A. No - Incorrect

Incorrect Answer Feedback: Incorrect because spreading rumors on social media to push up the stock price after buying the stock for his personal account reflects poorly on Chang's professional integrity. Therefore, Chang has violated both the Standard II(B), Market Manipulation and the Standard I(D), Misconduct.

B. Yes, the Standard relating to market manipulation only - Incorrect

Incorrect Answer Feedback: Incorrect because Standard I(D), Misconduct, addresses all conduct that reflects poorly on the professional integrity, good reputation, or competence of members and candidates. Any act that involves lying, cheating, stealing, or other dishonest conduct is a violation of this standard if the offense reflects adversely on a member's or candidate's professional activities. Spreading rumors on social media to push up the stock price reflects poorly on Chang's professional integrity. Therefore, Chang has also violated the Standard I(D), Misconduct.

C. Yes, both the Standard relating to market manipulation and the Standard relating to misconduct - Correct

Correct Answer Feedback: Correct because according to Standard II(B), Market Manipulation, information-based manipulation includes, but is not limited to, spreading false rumors to induce trading by others. For example, members and candidates must refrain from "pumping up" the price of an investment by issuing misleading positive information or overly optimistic projections of a security's worth only to later "dump" the investment (i.e., sell it) once the price, fueled by the misleading information's effect on other market participants, reaches an artificially high level. Chang spreads rumors on social media about a potential acquisition of AEC after buying the stock for his personal account. Therefore, Chang has violated the Standard II(B), Market Manipulation. In addition, Standard I(D), Misconduct, addresses all conduct that reflects poorly on the professional integrity, good reputation, or competence of members and candidates. Any act that involves lying, cheating, stealing, or other dishonest conduct is a violation of this standard if the offense reflects adversely on a member's or candidate's professional activities. Spreading rumors on social media to push up the stock price reflects poorly on Chang's professional integrity. Therefore, Chang has violated the Standard I(D), Misconduct.

Question 18

According to the Standards, which of the following information requires disclosure? • Information 1: A member's brother inherits a significant ownership of a stock that is part of the member's research coverage universe. The member has no beneficial ownership in her brother's investments. • Information 2: A member, a portfolio manager at an investment firm, does unpaid volunteer work during holiday weekends at a charity. • Information 3: Firm enters into an agreement with a mutual fund to receive commissions for recommending the fund to the firm's clients.

Correct answer: C

A. Information 1 - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VI(A), Avoid or Disclose Conflicts, members and candidates must disclose any materially beneficial ownership interest in a security or other investment that the member or candidate is recommending. The member has no beneficial ownership in her brother's investments. Therefore, Information 1 is not required to be disclosed.

B. Information 2 - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard VI(A), Avoid or Disclose Conflicts, situations that must be reported to employers include conflicts that would interfere with rendering unbiased investment advice and conflicts that would cause a member or candidate to act not in the employer's best interest. In this case, the member does unpaid volunteer work at a charity during holiday weekends which is her personal time. It does not have conflicts of interest to her employers. Therefore, Information 2 is not required to be disclosed.

C. Information 3 - Correct

Correct Answer Feedback: Correct because according to Standard VI(A), Avoid or Disclose Conflicts, disclosures must be made to clients regarding conflicts that may arise in fee arrangements, subadvisory agreements, or other situations involving nonstandard fee structures. Equally important is the disclosure of arrangements in which the firm benefits directly from investment recommendations. Therefore, Information 3 is required to be disclosed, because the firm benefits directly from its recommendation to invest in a certain mutual fund.

Question 19

According to the Standards, a member engages in information-based manipulation when she:

Correct answer: B

A. uses public information to implement trading strategies with an intent to exploit perceived market inefficiencies. - Incorrect

Incorrect Answer Feedback: Incorrect because the Standard relating to market manipulation is not intended to preclude transactions undertaken on legitimate trading strategies based on perceived market inefficiencies. Therefore, using public information to implement trading strategies to exploit perceived market inefficiencies is not an example of information-based manipulation.

B. induces other market participants to trade a security by issuing an overly optimistic projection of a security's value. - Correct

Correct Answer Feedback: Correct because according to the Standard relating to market manipulation, information-based manipulation includes spreading false rumors to induce trading by others. In this case, the member spreads a false rumor by issuing overly optimistic projections to induce other market participants to trade a security; therefore, the member engages in information-based manipulation.

C. relies on false rumors spread by other market participants to secure a dominant position in an underlying security. - Incorrect

Incorrect Answer Feedback: Incorrect because according to the Standard relating to market manipulation, information-based manipulation includes, but is not limited to, spreading false rumors to induce trading by others. The member relies on the false rumors existing in the market instead of spreading it. Therefore, she does not engage in information-based manipulation.

Question 20

According to the Standard relating to preservation of confidentiality, a member is required to maintain confidentiality of information communicated to her:

Correct answer: C

A. by her former clients only. - Incorrect

Incorrect Answer Feedback: Incorrect because Standard III(E), Preservation of Confidentiality requires that members and candidates preserve the confidentiality of information communicated to them by their clients, prospective clients, and former clients.

B. by her prospective clients only. - Incorrect

Incorrect Answer Feedback: Incorrect because Standard III(E), Preservation of Confidentiality requires that members and candidates preserve the confidentiality of information communicated to them by their clients, prospective clients, and former clients.

C. both by her former clients and by her prospective clients. - Correct

Correct Answer Feedback: Correct because Standard III(E), Preservation of Confidentiality requires that members and candidates preserve the confidentiality of information communicated to them by their clients, prospective clients, and former clients.

Question 21

Anisha Joshi, CFA, develops a product that selects mutual funds based on historical data. Joshi tests her methodology and produces simulated performance results. The promotional material for the product does not indicate that the results are simulated. Joshi has most likely violated the Standards relating:

Correct answer: B

A. only to fair dealing. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III(B), Fair Dealing, members and candidates must deal fairly and objectively with all clients when providing investment analysis, making investment recommendations, taking investment action, or engaging in other professional activities. There is nothing to suggest that Joshi has not dealt fairly and objectively with her clients. So, there is no violation of Standard III(B).

B. only to performance presentation. - Correct

Correct Answer Feedback: Correct because Standard III(D). Performance Presentation, requires members and candidates to provide credible performance information to clients and prospective clients and to avoid misstating performance or misleading clients and prospective clients about the investment performance of members or candidates or their firms. This standard encourages full disclosure of investment performance data to clients and prospective clients. By not disclosing that the results related to the product were simulated Joshi has violated the Standard.

C. both to fair dealing and to performance presentation. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard III(B), Fair Dealing, members and candidates must deal fairly and objectively with all clients when providing investment analysis, making investment recommendations, taking investment action, or engaging in other professional activities. There is nothing to suggest that Joshi has not dealt fairly and objectively with her clients. So, there is no violation of Standard III(B).

Question 22

Who is most likely responsible for claiming and maintaining compliance with the CFA Institute Global Investment Performance Standards (GIPS)?

Correct answer: A

A. The firm claiming compliance - Correct

Correct Answer Feedback: Correct because firms that claim compliance with the GIPS standards are responsible for their claim of compliance and for maintaining that compliance.

B. Independent verification firms - Incorrect

Incorrect Answer Feedback: Incorrect because an independent party is not responsible for the claim or for maintaining compliance. Independent third-party verifiers can be voluntarily hired by the firm claiming compliance to verify compliance.

C. The performance measurement department - Incorrect

Incorrect Answer Feedback: Incorrect because GIPS compliance is the responsibility of the firm claiming compliance and is not done on a department or on an individual basis.

Question 23

Mila Roy, CFA, heads the research division of an investment firm. Roy sends a recommended stock list to all of her firm's clients along with a notification that a detailed report on each stock is available on request. Roy later calls only her premium fee- paying clients and discusses her recommendations. Such additional services are available for all clients willing to pay premium fees and is fully disclosed as part of the client agreement. Has Roy violated the Standards?

Correct answer: A

A. No - Correct

Correct Answer Feedback: Correct because Standard III(B), Fair Dealing, requires CFA Institute members and candidates to deal fairly and objectively with all clients when providing investment analysis and making investment recommendations. It also requires that information about investment recommendations be disseminated in such a manner that all clients have a fair opportunity to act on the information. Roy distributes the recommended stock list to all clients simultaneously via email, giving each an opportunity to act on the information. Also, Roy is permitted to provide detailed reports to clients who pay a subscription fee for the service as long as (1) all clients are eligible to purchase the service and, (2) the different levels of service are disclosed to clients. Therefore, Roy has not violated Standard III(B). Further, according to Standard V(B), Communication with Clients and Prospective Clients, if recommendations are contained in capsule form (such as a recommended stock list), members and candidates should notify clients that additional information and analyses are available from the producer of the report. Roy sends a recommended stock list to all of her firm's clients along with a notification that a detailed report on each stock is available on request. Therefore, Roy has also not violated Standard V(B).

B. Yes, the Standard relating to fair dealing - Incorrect

Incorrect Answer Feedback: Incorrect because Standard III(B), Fair Dealing, requires CFA Institute members and candidates to deal fairly and objectively with all clients when providing investment analysis and making investment recommendations. It also requires that information about investment recommendations be disseminated in such a manner that all clients have a fair opportunity to act on the information. Roy distributes the recommended stock list to all clients simultaneously via email, giving each an opportunity to act on the information. Also, Roy is permitted to provide detailed reports to clients who pay a subscription fee for the service as long as (1) all clients are eligible to purchase the service and, (2) the different levels of service are disclosed to clients. Therefore, Roy has not violated Standard III(B).

C. Yes, the Standard relating to communication with clients and prospective clients - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard V(B), Communication with Clients and Prospective Clients, if recommendations are contained in capsule form (such as a recommended stock list), members and candidates should notify clients that additional information and analyses are available from the producer of the report. Roy sends a recommended stock list to all of her firm's clients along with a notification that a detailed report on each stock is available on request. Therefore, Roy has not violated Standard V(B).

Question 24

The GIPS standards were created to prevent the:

Correct answer: B

A. comparison of comparable time periods with other firms' results. - Incorrect

Incorrect Answer Feedback: Incorrect because the GIPS standards were created to allow for the comparison of consistent time periods to facilitate comparisons with other firms' results. It prevents misleading practices of varying time periods which are presenting performance for a selected time period during which the mandate produced excellent returns or out-performed its benchmarks - making comparison with other firms' results difficult or impossible.

B. selection of a top-performing portfolio to represent the firm's strategy. - Correct

Correct Answer Feedback: Correct because the GIPS standards were created to prevent misleading practices such as cherry-picking representative accounts which is the selection a top-performing portfolio to represent the firm's overall investment results for a specific mandate.

C. presentation of an average performance history that includes terminated portfolios. - Incorrect

Incorrect Answer Feedback: Incorrect because the GIPS standards were created to prevent survivorship bias which is presenting an 'average' performance history that excludes [not include] portfolios whose poor performance was weak enough to result in termination of the firm.

Question 25

According to the Standard relating to responsibilities of supervisors, a member who works as a department head managing large numbers of employees must:

Correct answer: A

A. have in-depth knowledge of the Code and Standards. - Correct

Correct Answer Feedback: Correct because according to Standard IV(C), Responsibilities of Supervisors, members and candidates acting as supervisors must also have in-depth knowledge of the Code and Standards.

B. personally evaluate the conduct of all employees in his department on a continuing basis. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard IV(C), Responsibilities of Supervisors, members and candidates with oversight responsibilities for large numbers of employees may not be able to personally evaluate the conduct of these employees on a continuing basis. These members and candidates may delegate supervisory duties to subordinates who directly oversee the other employees.

C. make reasonable efforts to ensure all employees at his firm comply with applicable laws and regulations. - Incorrect

Incorrect Answer Feedback: Incorrect because according to Standard IV(C), Responsibilities of Supervisors, members and candidates must make reasonable efforts to ensure that anyone subject to their supervision or authority complies with applicable laws, rules, regulations, and the Code and Standards. Therefore, the member is not required to make reasonable efforts to ensure all employees at his firm comply with applicable laws and regulations.

Question 26

Regarding challenges to ethical behavior, which of the following is not an example of a situational influence?

Correct answer: B

A. Bonuses - Incorrect

Incorrect Answer Feedback: Incorrect because this is an example of a situational influence. Bonuses, promotions, prestige, and loyalty to employer and colleagues are examples of situational influences that frequently have a disproportionate weight in our decision making.

B. Internal factors - Correct

Correct Answer Feedback: Correct because situational influences are external factors, such as environmental or cultural elements, that shape our thinking, decision making, and behavior. Therefore, internal decision making factors does not fall within the definition of situational influences.

C. Loyalty to employer - Incorrect

Incorrect Answer Feedback: Incorrect because this is an example of a situational influence. Bonuses, promotions, prestige, and loyalty to employer and colleagues are examples of situational influences that frequently have a disproportionate weight in our decision making.

Question 27

Mary Lawrence, CFA, is a research analyst. She archives her research notes for recommendation changes and discards her research notes for cases in which recommendations do not change. She also discards third-party research reports used to support her analysis. Lawrence has most likely violated the Standards:

Correct answer: C

A. only by discarding third-party research reports used to support her analysis. - Incorrect

Incorrect Answer Feedback: Incorrect as described in the response rationale of the correct answer.

B. only by discarding her research notes for cases in which recommendations do not change. - Incorrect

Incorrect Answer Feedback: Incorrect as described in the response rationale of the correct answer.

C. both by discarding third-party research reports used to support her analysis and by discarding her research notes for cases in which recommendations do not change. - Correct

Correct Answer Feedback: Correct because according to the Standard V(C), record retention, Members and candidates must retain records that substantiate the scope of their research and reasons for their actions or conclusions. The retention requirement applies to decisions to buy or sell a security as well as reviews undertaken that do not lead to a change in position. An example of supporting documentation that assists the member or candidate in meeting the requirements for retention are outside research reports.

Question 28

The central limit theorem:

Correct answer: B

A. requires that the population be approximately normally distributed. - Incorrect

Incorrect Answer Feedback: Incorrect because the central limit theorem allows us to make quite precise probability statements about the population mean by using the sample mean, whatever the distribution of the population (so long as it has finite variance), because the sample mean follows an approximate normal distribution for large- size samples. Therefore, having a normally distributed population is not a requirement for the central limit theorem to hold.

B. implies that the sample mean is a consistent estimator of the population mean. - Correct

Correct Answer Feedback: Correct because the central limit theorem states that the variance of the distribution of the sample mean is $\sigma 2/n$. The positive square root of variance is standard deviation. The standard deviation of a sample statistic is known as the standard error of the statistic. The sample mean, in addition to being an efficient

estimator, is also a consistent estimator of the population mean: As sample size n goes to infinity, its standard error, σ/√n, goes to 0 and its sampling distribution becomes concentrated right over the value of population mean, μ.

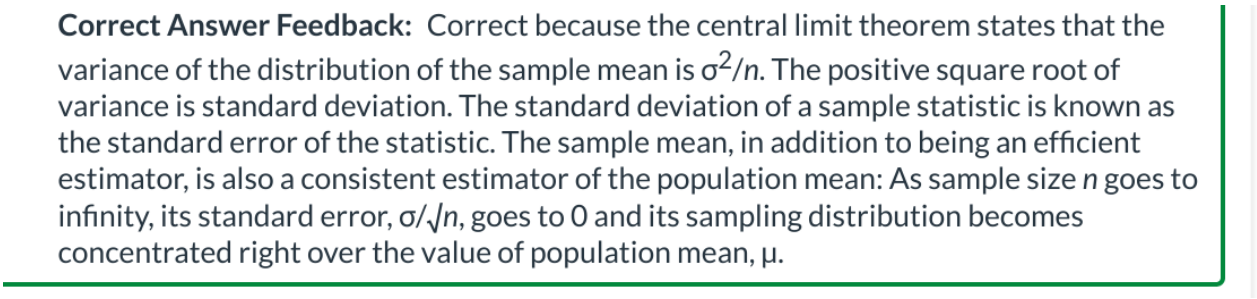


Figure 1: Source equation for question 28, option B

C. states that the product of independent random variables is normally distributed. - Incorrect

Incorrect Answer Feedback: Incorrect because the central limit theorem states that the sum and mean (not the product) of a large number of independent random variables is approximately normally distributed.

Question 29

A portfolio invested in two assets has an expected return of 11%. If expected returns for Assets A and B, respectively, are 8% and 12%, then the portfolio weight of Asset B is closest to:

Correct answer: C

A. 25%. - Incorrect

Incorrect Answer Feedback: Incorrect. This is the portfolio weight for Asset A. A weight of 25% for Asset B implies a portfolio return of 9%.

B. 50%. - Incorrect

Incorrect Answer Feedback: Incorrect. A weight of 50% for Asset B implies a portfolio return of 10%.

C. 75%. - Correct

Correct Answer Feedback: Correct. 11% = (w × 8%) + [(1 – w) × 12%], solving for w = 0.25. A A A Therefore w = 0.75 B (0.25 × 8%) + (0.75 × 12%) = 11%

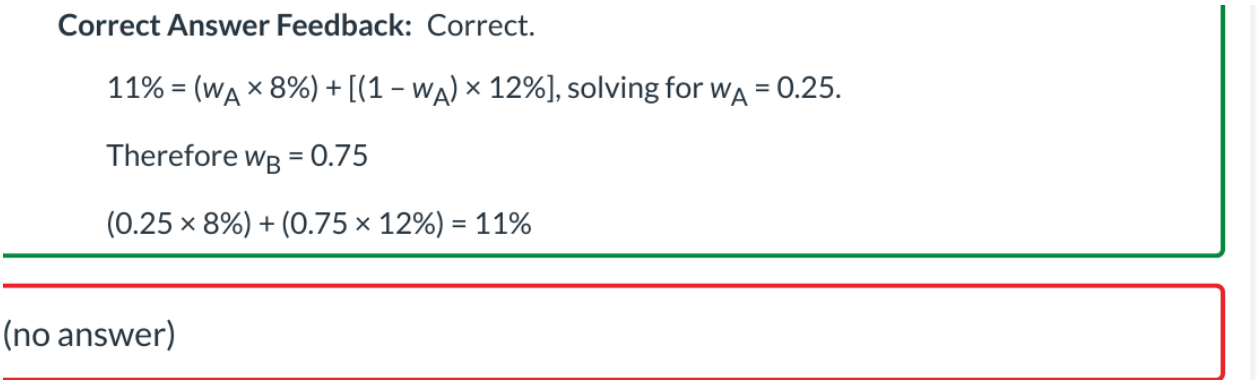


Figure 2: Source equation for question 29, option C

Question 30

In a test of independence based on contingency table data, the nonparametric test statistic to be used is:

Correct answer: C

A. F-distributed. - Incorrect

Incorrect Answer Feedback: Incorrect because if we want to test whether there is a relationship between the size and investment type (two classifications in a contingency table), we can perform a test of independence using a nonparametric test statistic that is chi-square distributed.

B. normally distributed. - Incorrect

Incorrect Answer Feedback: Incorrect because if we want to test whether there is a relationship between the size and investment type (two classifications in a contingency table), we can perform a test of independence using a nonparametric test statistic that is chi-square distributed.

C. chi-square distributed. - Correct

Correct Answer Feedback: Correct because if we want to test whether there is a relationship between the size and investment type (two classifications in a contingency table), we can perform a test of independence using a nonparametric test statistic that is chi-square distributed.

Question 31

An analyst gathers the following information (in £ millions) about a company:

Beginning accounts receivable	5,500
Ending accounts receivable	7,000
Revenue	35,000

Assuming all sales are made on credit, cash received from customers (in £ millions) is:

Correct answer: A

A. 33,500. - Correct

Correct Answer Feedback: Correct because to determine the approximate cash receipts from its customers, it is necessary to adjust this revenue amount by the net change in accounts receivable for the year. If accounts receivable increase during the year, revenue on an accrual basis is higher than cash receipts from customers, and vice versa. Cash received from customers = Beginning accounts receivable + Revenue – Ending accounts receivable. The calculation becomes 5,500 + 35,000 – 7,000 = 33,500.

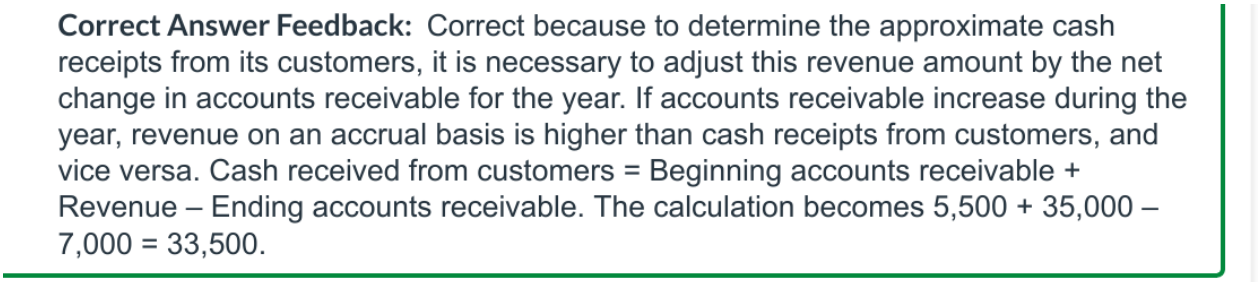


Figure 3: Source equation for question 31, option A

B. 35,000. - Incorrect

Incorrect Answer Feedback: Incorrect because it assumes that revenue received is all in cash. On a company's balance sheet, the item "trade receivables" or "accounts receivable" would indicate that a company provides credit to its customers.

C. 36,500. - Incorrect

Incorrect Answer Feedback: Incorrect because it reverses the Beginning and Ending accounts receivable in the calculation. The calculation becomes 7,000 + 35,000 – 5,500 = 36,500.

Incorrect Answer Feedback: Incorrect because it reverses the Beginning and Ending accounts receivable in the calculation. The calculation becomes 7,000 + 35,000 – 5,500 = 36,500.

Figure 4: Source equation for question 31, option C

Question 32

Double taxation of income occurs when a corporation pays taxes on its profits, and additional taxes are paid by the:

Correct answer: A

A. shareholders on dividends received. - Correct

Correct Answer Feedback: Correct because double taxation is when shareholders pay an additional tax on distributions (dividends) that are passed on to them.

B. corporation on distributions to shareholders. - Incorrect

Incorrect Answer Feedback: Incorrect because the second tax is paid by the shareholders, not the corporation, based on personal income (dividends) received.

C. shareholders on gains earned from their equity investment. - Incorrect

Incorrect Answer Feedback: Incorrect because the gains earned is not a form of income, but a change in the value of the ownership share.

Question 33

An analyst gathers the following fiscal year information (in ¥ millions) for a company:

Cash flow from operating activities	28,808
Purchase of equipment	1,300
Sale of equipment	1,750
Long-term debt repayment	470
Increase in short-term borrowing	930

FCFE is closest to:

Correct answer: C

A. ¥28,798 million. - Incorrect

Incorrect Answer Feedback: Incorrect because the calculation adds net debt repayment when it should subtract it. ¥28,798 = ¥28,808 – ¥1,300 + ¥1,750 – (–¥470 + ¥930)

Incorrect Answer Feedback: Incorrect because the calculation adds net debt repayment when it should subtract it.

¥28,798 = ¥28,808 – ¥1,300 + ¥1,750 – (–¥470 + ¥930)

Figure 5: Source equation for question 33, option A

B. ¥29,258 million. - Incorrect

Incorrect Answer Feedback: Incorrect because the calculation does not take into account the net debt repayment. ¥29,258 = ¥28,808 – ¥1,300 + ¥1,750.

C. ¥29,718 million. - Correct

Incorrect Answer Feedback: Incorrect because the calculation does not take into account the net debt repayment.

¥29,258 = ¥28,808 – ¥1,300 + ¥1,750.

Figure 6: Source equation for question 33, option B

Correct Answer Feedback: Correct because FCFE is the cash flow available to the company's common stockholders after all operating expenses and borrowing costs (principal and interest) have been paid and necessary investments in working capital and fixed capital have been made. FCFE can be computed as

FCFE = CFO – FCInv + net borrowing

Where:

CFO = Cash flow from operating activities

FCInv = Capital expenditures (fixed capital, such as equipment)

FCFE = ¥28,808 – ¥1,300 + ¥1,750 – ¥470 + ¥930 = ¥29,718.

Figure 7: Source equation for question 33, option C

(no answer)

Figure 8: Source equation for question 33, option C

Question 34

A positive movement in a lagging indicator would least likely be used to:

Correct answer: B

A. identify a past condition of the economy. - Incorrect

Incorrect Answer Feedback: Incorrect because a positive move in a lagging indicator is most likely identifying an upturn in economic activity that occurred in the past.

B. identify an expected future economic upturn. - Correct

Correct Answer Feedback: Correct because a positive movement in a lagging indicator would most likely be used to confirm that an existing expansion is underway or has already occurred. Only a leading indicator would help identify or predict a future economic event.

C. confirm that an expansion is currently underway. - Incorrect

Incorrect Answer Feedback: Incorrect because a positive move in a lagging indicator by itself is insufficient to indicate a positive expansion. However, confirmation would be required from positive changes in a coincident indicator to indicate expansion.

Question 35

Which of the following is an example of cooperative country behavior?

Correct answer: C

A. Autarky - Incorrect

Incorrect Answer Feedback: Incorrect because autarky describes countries seeking political self-sufficiency with little or no external trade or finance.

B. Hegemony - Incorrect

Incorrect Answer Feedback: Incorrect because hegemonic countries tend to be regional or even global leaders, and they use their political or economic influence of others to control resources. State-owned enterprises tend to control key export markets.

C. Bilateralism - Correct

Correct Answer Feedback: Correct because bilateralism is the conduct of political, economic, financial, or cultural cooperation between two countries.

Question 36

A firm is operating under an oligopoly when it is:

Correct answer: C

A. the sole provider of a good or service. - Incorrect

Incorrect Answer Feedback: Incorrect because a single provider of a good or service is representative of a monopoly, not an oligopoly. In pure monopoly markets, there are no other good substitutes for the given product or service. There is a single seller, which, if allowed to operate without constraint, exercises considerable power over pricing and output decisions.

B. one of many providers of a good or service. - Incorrect

Incorrect Answer Feedback: Incorrect because one or many providers of a good or service is representative of both monopolistic competition and perfect competition, not an oligopoly. Monopolistic competition is also highly competitive. The competitive characteristic is a notably large number of firms. Similarly, perfect competition is a reality —for example, in several commodities markets, where sellers and buyers have a strictly homogeneous product and no single producer is large enough to influence market prices. Product differentiation is not possible.

C. one of a small number of providers of a good or service. - Correct

Correct Answer Feedback: Correct because the oligopoly market structure is based on a relatively small number of firms supplying the market.

Question 37

An analyst gathers the following information (in € millions) about a company's machine:

Carrying amount prior to impairment	8.0
Present value of expected future cash flows	7.4
Undiscounted expected future cash flows	7.8
Costs to sell	0.2
Impairment loss	0.5

The fair value of the machine (in € millions) is:

Correct answer: B

A. 7.5. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer is based on US GAAP, which measures impairment loss as the difference between the asset's fair value and carrying amount if the carrying amount of the asset is considered not recoverable. The carrying amount of an asset 'group' is considered not recoverable when it exceeds the undiscounted expected future cash flows of the group. According to US GAAP, Fair value of the asset = Carrying amount – Impairment loss = €(8.0 – 0.5) million = €7.5 million.

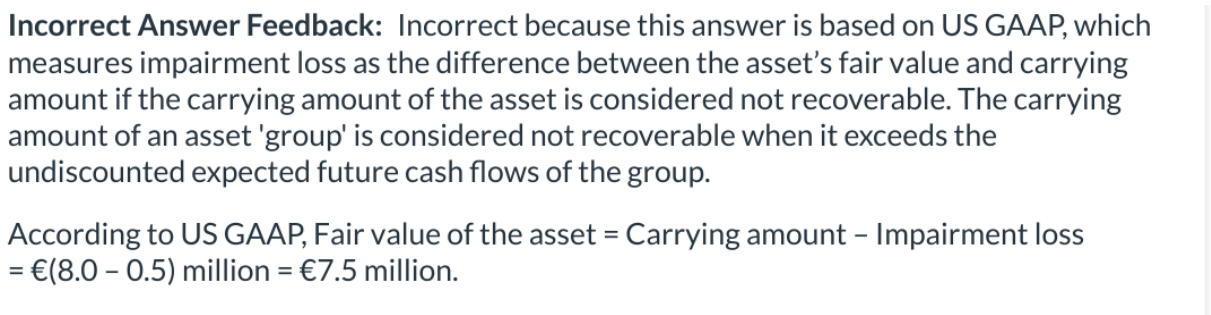


Figure 9: Source equation for question 37, option A

B. 7.7. - Correct

Correct Answer Feedback: Correct because under IFRS, an asset's recoverable amount is the higher of its fair value less costs to sell and its value in use. Expressing this in the form of an equation: Recoverable amount = max[(Fair value – Costs to sell); Value in use] = max[(Fair value – €0.2 million); €7.4 million]. Given that under IAS 36, an impairment loss is measured as the excess of carrying amount over the recoverable amount of the asset, the recoverable amount can also be calculated as carrying value less impairment loss. Expressing this as an equation: Recoverable amount = Carrying value – Impairment loss = €(8.0 – 0.5) million = €7.5 million Equating equations 1 & 2: max[(Fair value – €0.2 million); €7.4 million] = €7.5 million Fair value = €7.5 million + €0.2 million = €7.7 million.

Correct Answer Feedback: Correct because under IFRS, an asset's recoverable amount is the higher of its fair value less costs to sell and its value in use. Expressing this in the form of an equation: Recoverable amount = max[(Fair value – Costs to sell); Value in use] = max[(Fair value – €0.2 million); €7.4 million].

Given that under IAS 36, an impairment loss is measured as the excess of carrying amount over the recoverable amount of the asset, the recoverable amount can also be calculated as carrying value less impairment loss. Expressing this as an equation:

Recoverable amount = Carrying value – Impairment loss = €(8.0 – 0.5) million = €7.5 million

Equating equations 1 & 2: max[(Fair value – €0.2 million); €7.4 million] = €7.5 million

Fair value = €7.5 million + €0.2 million = €7.7 million.

Figure 10: Source equation for question 37, option B

C. 7.8. - Incorrect

Incorrect Answer Feedback: Incorrect because this answer assumes that fair value is the carrying amount less costs to sell = €(8.0 – 0.2) million = €7.8 million. Instead, under IFRS, an impairment loss is measured as the excess of carrying amount over the recoverable amount of the asset, while the asset's recoverable amount is the higher of its fair value less costs to sell and its value in use.

Incorrect Answer Feedback: Incorrect because this answer assumes that fair value is the carrying amount less costs to sell = €(8.0 – 0.2) million = €7.8 million. Instead, under IFRS, an impairment loss is measured as the excess of carrying amount over the recoverable amount of the asset, while the asset's recoverable amount is the higher of its fair value less costs to sell and its value in use.

Figure 11: Source equation for question 37, option C

(no answer)

Figure 12: Source equation for question 37, option C

Question 38

An advantage of indirect taxes as a fiscal policy tool is that such taxes:

Correct answer: A

A. can be adjusted almost immediately. - Correct

Correct Answer Feedback: Correct because indirect taxes, which are taxes on spending including excise taxes on fuel, alcohol, and tobacco as well as taxes on gambling winnings, can be adjusted almost immediately after they are announced and can influence spending behavior instantly.

B. minimize interference with consumer choices. - Incorrect

Incorrect Answer Feedback: Incorrect because indirect taxes support social policies that are specifically aimed at influencing and deterring consumer choices related to such consumption and activities.

C. have a greater impact on aggregate spending and output than direct government spending. - Incorrect

Incorrect Answer Feedback: Incorrect because direct government spending has a far bigger impact on aggregate spending and output than tax cuts or transfer increases.

Question 39

The notes to the financial statements are required to provide:

Correct answer: B

A. explanatory information about every line item on the balance sheet and income statement. - Incorrect

Incorrect Answer Feedback: Incorrect because it is not a requirement to provide information about every line item and for many companies, the financial notes and supplemental schedules provide explanatory information about every line item (or almost every line item) on the balance sheet and income statement.

B. information about the accounting policies, methods, and estimates used to prepare the financial statements. - Correct

Correct Answer Feedback: Correct because the notes disclose information about the accounting policies, methods, and estimates used to prepare the financial statements.

C. management comments over any favorable or unfavorable trends, significant events and uncertainties that may affect the company. - Incorrect

Incorrect Answer Feedback: Incorrect because management highlights are covered by the management commentary or management's discussion and analysis (MD&A) and not the financial notes. Also, in the United States, the SEC requires listed companies to provide an MD&A and specifies the content. Management must highlight any favorable or unfavorable trends and identify significant events and uncertainties that affect the company's liquidity, capital resources, and results of operations.

Question 40

At which stage in its life cycle would a typical company most likely have more debt than equity in its capital structure?

Correct answer: C

A. Start-up stage - Incorrect

Incorrect Answer Feedback: Incorrect because at this early stage, debt capital is typically not available or available only at a high cost. From the perspective of an early- stage issuer, debt may be an attractive way to reduce or avoid the dilution associated with equity issuance. However, with cash flows that are negative and unpredictable, a typical start-up would have difficulty making regular debt payments, or 'servicing the debt.' Even if debt is available, the cost, inflexibility, and risk associated with borrowing are often unattractive to a start-up. As a result, debt is a negligible component of the capital structures of most start-up companies.

B. Growth stage - Incorrect

Incorrect Answer Feedback: Incorrect because as a company exits the growth stage and progresses through the growth stage, debt financing might become more accessible, but the company might want to use debt conservatively and continue relying primarily on equity financing. Both the availability and the terms of debt financing improve for the company during this stage. Many growth companies use debt conservatively in order to preserve operational and financial flexibility and minimize the risk of financial distress. Equity remains the predominant source of capital.

C. Mature stage - Correct

Correct Answer Feedback: Correct because at the mature stage, the company becomes able to support low-cost debt, often on an unsecured basis. From the company's perspective, debt financing is likely to be more attractive than higher-cost equity financing. Also, as companies mature, business risk typically declines, and their cash

flows turn positive and become increasingly predictable, allowing for greater use of leverage on more attractive (less costly) financing terms. Debt then becomes a larger component of their capital structures.

Question 41

Business models should have:

Correct answer: C

A. a value chain only. - Incorrect

Incorrect Answer Feedback: Incorrect because key questions to better ascertain a business model are: “What is the firm’s value proposition to its target customer(s)?” and “What is the firm’s value chain?” Thus, both, value proposition and value chain are components or aspects of a firm’s business model.

B. a value proposition only. - Incorrect

Incorrect Answer Feedback: Incorrect because key questions to better ascertain a business model are: “What is the firm’s value proposition to its target customer(s)?” and “What is the firm’s value chain?” Thus, both, value proposition and value chain are components or aspects of a firm’s business model.

C. both a value chain and a value proposition. - Correct

Correct Answer Feedback: Correct because key questions to better ascertain a business model are: “What is the firm’s value proposition to its target customer(s)?” and “What is the firm’s value chain?” Thus, both, value proposition and value chain are components or aspects of a firm’s business model.

Question 42

An analyst gathers the following information about a company’s capital investment:

Initial cash outlay	$90 million
Annual before-tax cash flows (year-end) for Year 1 through Year 6	$50 million
Marginal tax rate	15%
Required rate of return	12%

The net present value of the investment is closest to:

Correct answer: A

A. $85 million. - Correct

Correct Answer Feedback: Correct because for a capital investment with one investment outlay, made initially, the net present value (NPV) is the present value of the future after-tax cash flows minus the investment outlay NPV = nΣCF /(1 + r)t − t=1 t Outlay We calculate the present value of the future after-tax cash flows using the calculator with the following parameters: Yearly after-tax payment/cash-flow: PMT = 50 × (1 – 0.15) = 42.5 Number of payments: N = 6 Int. rate = 12%, To calculate the NPV, from the present value we subtract the initial $90 million outflow. Accordingly, NPV = $174.7 million – $90.0 million = $84.7 ≈ $85 million.

B. $116 million. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses the before-tax cash flow of $50 million in the calculation (instead of the after-tax cash flow of $42.5 million). Calculating the present value of the cash flows with parameters PMT = 50, N = 6 and Int. Rate = 12% gives a present value of $205.57 million, and subtracting the initial $90 million outflow, the NPV is incorrectly calculated as NPV = $205.57 million – $90 million = $115.6 million ≈ $116 million.

Correct Answer Feedback: Correct because for a capital investment with one investment outlay, made initially, the net present value (NPV) is the present value of the future after-tax cash flows minus the investment outlay NPV = $_{t=1}^{n}\Sigma CF_t/(1+r)^t$ – Outlay

We calculate the present value of the future after-tax cash flows using the calculator with the following parameters:

Yearly after-tax payment/cash-flow: PMT = 50 × (1 – 0.15) = 42.5

Number of payments: N = 6

Int. rate = 12%,

To calculate the NPV, from the present value we subtract the initial $90 million outflow. Accordingly, NPV = $174.7 million – $90.0 million = $84.7 ≈ $85 million.

Figure 13: Source equation for question 42, option A

Incorrect Answer Feedback: Incorrect because it uses the before-tax cash flow of $50 million in the calculation (instead of the after-tax cash flow of $42.5 million). Calculating the present value of the cash flows with parameters PMT = 50, N = 6 and Int. Rate = 12% gives a present value of $205.57 million, and subtracting the initial $90 million outflow, the NPV is incorrectly calculated as NPV = $205.57 million – $90 million = $115.6 million ≈ $116 million.

Figure 14: Source equation for question 42, option B

C. $175 million. - Incorrect

Incorrect Answer Feedback: Incorrect because it omits the initial cash outlay. Accordingly, the NPV = $174.7 ≈ $175 million.

Incorrect Answer Feedback: Incorrect because it omits the initial cash outlay. Accordingly, the NPV = $174.7 ≈ $175 million.

Figure 15: Source equation for question 42, option C

Question 43

An analyst is preparing financial projections for a company held in her firm's equity portfolio. When determining an appropriate time horizon for her company projections, the analyst should consider:

Correct answer: C

A. the portfolio's turnover only. - Incorrect

Incorrect Answer Feedback: Incorrect because the cyclicality of the industry could also influence the analyst's choice of time frame because the forecast period should be long enough to allow the business to reach an expected mid-cycle level of sales and profitability.

B. cyclicality of the company's industry only. - Incorrect

Incorrect Answer Feedback: Incorrect because the time frame should ideally correspond with average portfolio turnover of the portfolio.

C. both the portfolio's turnover and the cyclicality of the company's industry. - Correct

Correct Answer Feedback: Correct because the time frame should ideally correspond with average portfolio turnover of the portfolio. Furthermore, the cyclicality of the industry could also influence the analyst's choice of time frame because the forecast period should be long enough to allow the business to reach an expected mid-cycle level of sales and profitability.

Question 44

An analyst gathers the following information about a company's capital structure, cost of capital and marginal tax rate:

Market value of debt	$1,000 million
Market value of equity	$500 million
Book value of debt	$900 million
Book value of equity	$300 million
Pre-tax cost of debt	4%
Cost of equity	6%
Marginal tax rate	30%

If interest is fully tax deductible, the WACC is closest to:

Correct answer: B

A. 3.6%. - Incorrect

Incorrect Answer Feedback: Incorrect because this distractor incorrectly uses book values instead of market values to calculate weights. WACC = w r (1 – t) + w r + d d p p w r ≠ (900/(900 + 300)) × 0.04 × (1 – 0.30) + (300/(900+300)) × 0.06 = 0.036 = e e 3.6%.

Incorrect Answer Feedback: Incorrect because this distractor incorrectly uses book values instead of market values to calculate weights. WACC = $w_d r_d(1 - t) + w_p r_p + w_e r_e \neq (900/(900 + 300)) \times 0.04 \times (1 - 0.30) + (300/(900+300)) \times 0.06 = 0.036 = 3.6\%$.

Figure 16: Source equation for question 44, option A

Figure 17: Source equation for question 44, option A

B. 3.9%. - Correct

Correct Answer Feedback: Correct because WACC = w r (1 – t) + w r + w r d d p p e e, where w is the proportion of debt, d r is the before-tax marginal cost of debt, d w is the proportion of preferred stock, p r is the marginal cost of preferred stock, p w is the proportion of equity, and e r is the marginal cost of equity. e Therefore WACC = (1000/(1000 + 500)) × (0.04)(1 – 0.30) + (500/(1000 + 500)) × 0.06 = 0.03867 ≈ 3.9%.

Correct Answer Feedback: Correct because WACC = $w_d r_d(1 - t) + w_p r_p + w_e r_e$,

where

w_d is the proportion of debt,

r_d is the before-tax marginal cost of debt,

w_p is the proportion of preferred stock,

r_p is the marginal cost of preferred stock,

w_e is the proportion of equity, and

r_e is the marginal cost of equity.

Therefore WACC = $(1000/(1000 + 500)) \times (0.04)(1 - 0.30) + (500/(1000 + 500)) \times 0.06 = 0.03867 \approx 3.9\%$.

Figure 18: Source equation for question 44, option B

C. 4.7%. - Incorrect

Incorrect Answer Feedback: Incorrect because this response does not take into account the tax deductibility of interest, i.e. WACC = w r (1 – t) + w r + w r ≠ d d p p e e (1000/(1000+500)) × 0.04 + (500/(1000+500)) × 0.06 = 0.04667 ≈ 4.7%.

Incorrect Answer Feedback: Incorrect because this response does not take into account the tax deductibility of interest, i.e. WACC = $w_d r_d(1 - t) + w_p r_p + w_e r_e \neq (1000/(1000+500)) \times 0.04 + (500/(1000+500)) \times 0.06 = 0.04667 \approx 4.7\%$.

Figure 19: Source equation for question 44, option C

Question 45

The regulation of economic activity over time is associated with:

Correct answer: C

A. fiscal policy only. - Incorrect

Incorrect Answer Feedback: Incorrect because both monetary and fiscal policies are used to regulate economic activity over time.

B. monetary policy only. - Incorrect

Incorrect Answer Feedback: Incorrect because both monetary and fiscal policies are used to regulate economic activity over time.

C. both fiscal policy and monetary policy. - Correct

Correct Answer Feedback: Correct because both monetary and fiscal policies are used to regulate economic activity over time.

Question 46

An analyst gathers the following one-period continuously compounded returns.

r 0,1	4.2%
r 1,2	3.0%
r 2,3	–1.0%

The continuously compounded return from time t = 0 to time t = 3 is closest to:

Correct answer: B

A. 6.07%. - Incorrect

Incorrect Answer Feedback: Incorrect because the candidate mistakenly assumes the values in the table are holding period returns and then uses the formula r = ln(1 + t,t+1 R) to convert them to one-period continuously compounded returns. Then, these are t,t+1 added to calculate the continuously compounded return to time T as r = r + r + 0,3 2,3 1,2 r ≠ ln(1 – 1.0%) + ln(1 + 3.0%) + ln(1 + 4.2%) = –0.01005 + 0.02956 + 0.04114 = 0,1 0.06065 ≈ 6.07%.

Incorrect Answer Feedback: Incorrect because the candidate mistakenly assumes the values in the table are holding period returns and then uses the formula $r_{t,t+1} = \ln(1 + R_{t,t+1})$ to convert them to one-period continuously compounded returns. Then, these are added to calculate the continuously compounded return to time T as $r_{0,3} = r_{2,3} + r_{1,2} + r_{0,1} \neq \ln(1 - 1.0\%) + \ln(1 + 3.0\%) + \ln(1 + 4.2\%) = -0.01005 + 0.02956 + 0.04114 = 0.06065 \approx 6.07\%$.

Figure 20: Source equation for question 46, option A

B. 6.20%. - Correct

Correct Answer Feedback: Correct because the continuously compounded return to time T is the sum of the one-period continuously compounded returns: r = r + 0,T T−1,T r + . . . + r . In this case, r = r + r + r = –1.0% + 3.0% + 4.2% = 6.20%. T−2,T−1 0,1 0,3 2,3 1,2 0,1

Correct Answer Feedback: Correct because the continuously compounded return to time T is the sum of the one-period continuously compounded returns: $r_{0,T} = r_{T-1,T} + r_{T-2,T-1} + \ldots + r_{0,1}$. In this case, $r_{0,3} = r_{2,3} + r_{1,2} + r_{0,1} = -1.0\% + 3.0\% + 4.2\% = 6.20\%$.

Figure 21: Source equation for question 46, option B

C. 6.25%. - Incorrect

Incorrect Answer Feedback: Incorrect because the candidate does calculations for holding period returns. i.e r = {(1 + r) × (1 + r) × (1 + r)} – 1 = {(1 + 4.2%) × (1 + 0,3 0,1 2,3 1,2 3.0%) × (1 – 1.0%)} – 1 = (1.042 × 1.03 × 0.99) – 1 = 1.06253 – 1 = 0.06253 ≈ 6.25%.

Incorrect Answer Feedback: Incorrect because the candidate does calculations for holding period returns. i.e $r_{0,3} = \{(1 + r_{0,1}) \times (1 + r_{2,3}) \times (1 + r_{1,2})\} - 1 = \{(1 + 4.2\%) \times (1 + 3.0\%) \times (1 - 1.0\%)\} - 1 = (1.042 \times 1.03 \times 0.99) - 1 = 1.06253 - 1 = 0.06253 \approx 6.25\%$.

Figure 22: Source equation for question 46, option C

Question 47

An analyst runs a simple linear regression to test whether the variation in the demand for corn explains the variation in the supply of wheat. In this model, the demand for corn is a(n):

Correct answer: C

A. indicator variable. - Incorrect

Incorrect Answer Feedback: Incorrect because demand for corn is not limited to values of 0 or 1. Suppose we want to examine whether a company's quarterly earnings announcements influence its monthly stock returns. In this case, we could use an indicator variable, or dummy variable, that takes on only the values 0 or 1 as the independent variable.

B. explained variable. - Incorrect

Incorrect Answer Feedback: Incorrect because variation in the demand for corn is being used to explain the variation in the supply of wheat. Therefore the variation in the supply of wheat is the dependent variable, or explained variable. We refer to the variable whose variation is being explained as the dependent variable, or the explained variable; it is typically denoted by Y.

C. independent variable. - Correct

Correct Answer Feedback: Correct because the variation in the demand for corn is being used to explain the variation in the supply of wheat. We refer to the variable whose variation is being used to explain the variation of the dependent variable as the independent variable, or the explanatory variable; it is typically denoted by X.

Question 48

A company must disclose for each class of long-lived assets the amounts of impairment losses recognized in the period and where those losses are recognized on the financial statements under:

Correct answer: C

A. IFRS only. - Incorrect

Incorrect Answer Feedback: Incorrect because this is required under both IFRS and US GAAP.

B. US GAAP only. - Incorrect

Incorrect Answer Feedback: Incorrect because this is required under both IFRS and US GAAP.

C. both IFRS and US GAAP. - Correct

Correct Answer Feedback: Correct because under IFRS, a company must disclose for each class of assets the amounts of impairment losses and reversals of impairment losses recognized in the period and where those are recognized on the financial statements. The company must also disclose in aggregate the main classes of assets affected by impairment losses and reversals of impairment losses and the main events and circumstances leading to recognition of these impairment losses and reversals of impairment losses. Under US GAAP, there is no reversal of impairment losses for assets held for use. The company must disclose a description of the impaired asset, what led to the impairment, the method of determining fair value, the amount of the impairment loss, and where the loss is recognized on the financial statements.

Question 49

A country in fiscal balance with a trade surplus will most likely:

Correct answer: C

A. sell assets to foreigners to reduce the imbalance. - Incorrect

Incorrect Answer Feedback: Incorrect because a country with a trade surplus will buy assets from foreigners to reduce the imbalance.

B. increase the imbalance by lending to foreign countries. - Incorrect

Incorrect Answer Feedback: Incorrect because a country with a trade surplus will decrease the imbalance by lending to foreign countries.

C. have an excess of domestic saving relative to investment spending. - Correct

Correct Answer Feedback: Correct because trade surplus arises when exports exceed imports—that is, X > M. A fiscal balance arises when government spending equals taxation—that is, G = T. Since X – M = (S – I) + (T – G), if T = G, then a trade surplus can arise only when there is an excess of domestic saving over investment spending.

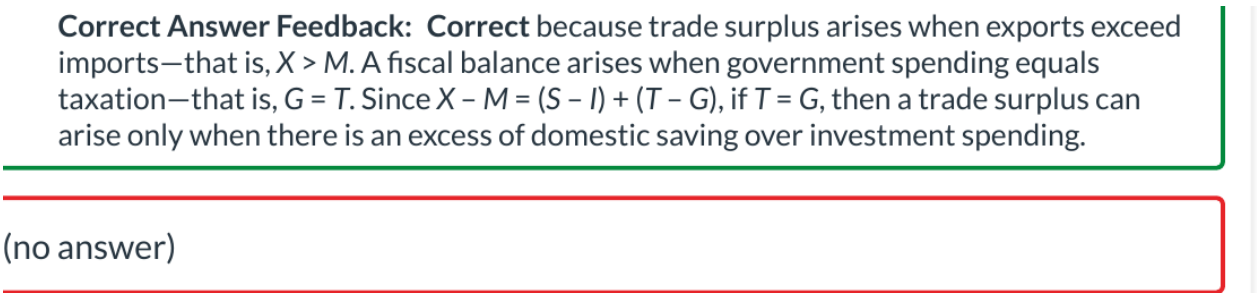


Figure 23: Source equation for question 49, option C

Question 50

Which of the following changes to compensation packages would most likely result in management becoming more risk averse in its corporate decision making?

Correct answer: B

A. Decreasing the length of management tenure - Incorrect

Incorrect Answer Feedback: Incorrect because when the overall level of board director or manager compensation, or tenure, is excessive, the result may lead to the avoidance of risk motivated by a vested interest in keeping one's position. Thus increasing, not decreasing, management tenure could result in more risk averse behavior by management.

B. Eliminating stock grants and options from management compensation - Correct

Correct Answer Feedback: Correct because a compensation package relying too heavily on stock grants and options can motivate risk-taking behavior by management, since option holders participate only in upside share price moves. Similarly, too little or no use of stock grants and options awarded to managers can lead to the opposite result. However, managers and directors without a meaningful equity stake in the company are typically more risk averse in their corporate decision making so they can better protect their long-term engagement by the company.

C. Tying management compensation to the size of the company's business - Incorrect

Incorrect Answer Feedback: Incorrect because tying management compensation to the size of the company's business would increase the risk-taking behavior of management. When director and management compensation are high and tied to the size of the business, it can also lead to 'growth for growth's sake' in which managers are motivated to pursue acquisitions and expansion that might not increase shareholder value.

Question 51

Members of a company's board of directors are most likely required to:

Correct answer: B

A. report to the company's CEO. - Incorrect

Incorrect Answer Feedback: Incorrect because the board is also responsible for hiring the CEO. The CEO reports to the board, the board does not report to the CEO.

B. display loyalty to the company. - Correct

Correct Answer Feedback: Correct because directors are required under law to display a high standard of prudence, care, and loyalty to the company.

C. work for the company or the industry in which the company operates. - Incorrect

Incorrect Answer Feedback: Incorrect because most codes addressing corporate governance standards require that board members represent a diverse mix of expertise, backgrounds, and competencies, with best practice generally dictating at least one third of the board be independent.

Question 52

An analyst gathers the following information about a company's common stock In Year 1: 1 January: 200,000 shares outstanding 1 June: 50,000 shares issued 1 August: two-for-one stock split 31 December: 500,000 shares outstanding To calculate earnings per share for Year 1, the company's weighted average number of shares outstanding is closest to:

Correct answer: C

A. 350,000. - Incorrect

Incorrect Answer Feedback: Incorrect. This response is a simple average of beginning and ending shares outstanding: (200,000 + 500,000)/2 = 350,000.

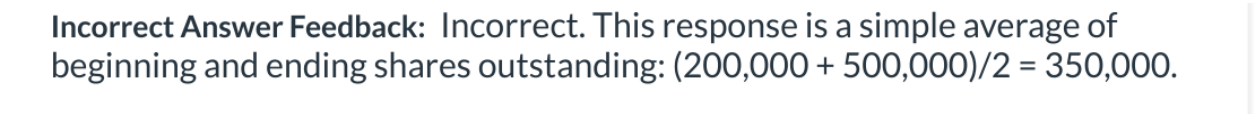


Figure 24: Source equation for question 52, option A

B. 375,000. - Incorrect

Incorrect Answer Feedback: Incorrect. This answer does not account for the share split for the weighted average shares from January to May. Event WACS 200,000 × 5/12 (January–May) 83,333 250,000 × 7/12 (June–December) 145,833 2:1 split for the June–Dec shares 291,667 Sum of lines 1 and 3 375,000

C. 458,332. - Correct

Correct Answer Feedback: Correct. The weighted average number of common shares (WACS) is determined by the length of time each quantity of shares was outstanding. A stock split is treated as if it occurred at the beginning of the year. Event WACS 200,000 × 5/12 (January–May) 83,333 250,000 × 7/12 (June–December) 145,833 Total before split 229,166 Including effect of 2:1 split 458,332

Question 53

The most appropriate treatment of goodwill is that it be:

Correct answer: B

Incorrect Answer Feedback: Incorrect. This answer does not account for the share split for the weighted average shares from January to May.

Event	WACS
200,000 × 5/12 (January–May)	83,333
250,000 × 7/12 (June–December)	145,833
2:1 split for the June–Dec shares	291,667
Sum of lines 1 and 3	375,000

Figure 25: Source equation for question 52, option B

Figure 26: Source equation for question 52, option B

Correct Answer Feedback: Correct. The weighted average number of common shares (WACS) is determined by the length of time each quantity of shares was outstanding. A stock split is treated as if it occurred at the beginning of the year.

Event	WACS
200,000 × 5/12 (January–May)	83,333
250,000 × 7/12 (June–December)	145,833
Total before split	229,166
Including effect of 2:1 split	458,332

Figure 27: Source equation for question 52, option C

A. amortized. - Incorrect

Incorrect Answer Feedback: Incorrect because goodwill is not amortized.

B. tested annually for impairment. - Correct

Correct Answer Feedback: Correct because goodwill is tested for impairment annually.

C. fully expensed at the time of acquisition. - Incorrect

Incorrect Answer Feedback: Incorrect because goodwill is capitalized, rather than expensed.

Question 54

A corporate takeover mechanism by which shareholders are persuaded to vote for a group seeking a controlling position on a company's board of directors best defines a:

Correct answer: B

A. tender offer. - Incorrect

Incorrect Answer Feedback: Incorrect because corporate takeovers can be pursued in several different ways, Managerial teams can also be displaced through a tender offer, which involves shareholders selling their interests directly to the group seeking to gain control. A takeover mechanism by which shareholders are persuaded to vote for a group seeking a controlling position on a company's board of directors describes a proxy contest and not a tender offer.

B. proxy contest. - Correct

Correct Answer Feedback: Correct because corporate takeovers can be pursued in several different ways. One mechanism is the proxy contest (or proxy fight). In a proxy contest, shareholders are persuaded to vote for a group seeking a controlling position on a company's board of directors.

C. hostile takeover. - Incorrect

Incorrect Answer Feedback: Incorrect because corporate takeovers can be pursued in several different ways a hostile takeover is an attempt by one entity to acquire a company without the consent of the company's management. A takeover mechanism by which shareholders are persuaded to vote for a group seeking a controlling position on a company's board of directors describes a proxy contest and not a hostile takeover.

Question 55

If all observations in a dataset have distinct positive values, which of the following relationships holds?

Correct answer: C

A. Arithmetic mean < Harmonic mean < Geometric mean - Incorrect

Incorrect Answer Feedback: Incorrect because a mathematical fact concerning the harmonic, geometric, and arithmetic means is that unless all the observations in a dataset have the same value, the harmonic mean is less than the geometric mean, which, in turn, is less than the arithmetic mean.

B. Geometric mean < Arithmetic mean < Harmonic mean - Incorrect

Incorrect Answer Feedback: Incorrect because a mathematical fact concerning the harmonic, geometric, and arithmetic means is that unless all the observations in a dataset have the same value, the harmonic mean is less than the geometric mean, which, in turn, is less than the arithmetic mean.

C. Harmonic mean < Geometric mean < Arithmetic mean - Correct

Correct Answer Feedback: Correct because a mathematical fact concerning the harmonic, geometric, and arithmetic means is that unless all the observations in a dataset have the same value, the harmonic mean is less than the geometric mean, which, in turn, is less than the arithmetic mean.

Question 56

An analyst gathers the following information about a company:

Debt-to-equity ratio based on market value	43%
Debt-to-equity ratio based on book value	52%

The weight of debt the analyst should use when determining the company's target capital structure is closest to:

Correct answer: A

A. 30%. - Correct

Correct Answer Feedback: Correct because individuals outside the company, however, such as analysts, typically do not know the target capital structure and must estimate it using one of several methods. Assume the company's current capital structure, at market value weights for the components, represents the company's target capital structure. A simple way of transforming a debt-to-equity ratio (D/E) into a weight—that is, D/(D + E) —is to divide D/E by 1 + D/E. Thus the analyst should use 0.43 / (1 + 0.43) = 30.07% ~ 30% as the weight of debt in any estimate of the company's target capital structure.

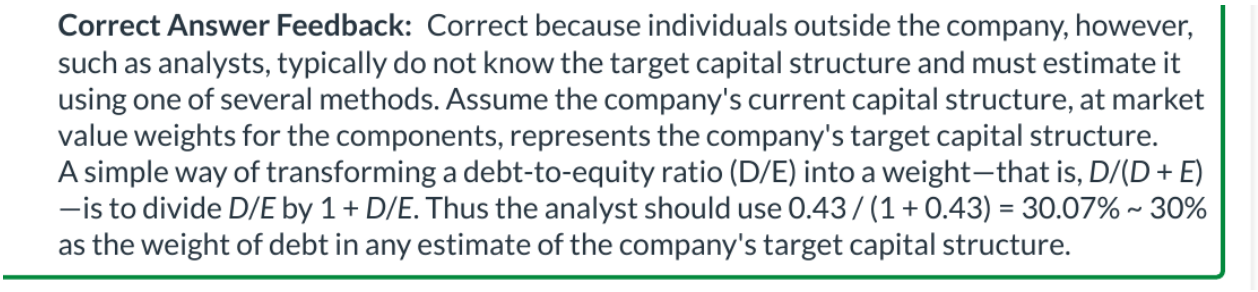


Figure 28: Source equation for question 56, option A

B. 43%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the debt-to-equity ratio based on market value, not the weight of debt based on market value.

C. 52%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the debt-to-equity ratio based on book value, not the weight of debt based on market value.

Question 57

A portfolio has two stocks with equal weighting. The variance of returns for each stock is 100 percent squared, and the covariance is 50 percent squared. The portfolio standard deviation of returns is closest to:

Correct answer: B

A. 7.9%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the portfolio standard deviation calculated from an incorrect formula for portfolio variance, where the "2" is omitted from the covariance term, i.e. σ2(R) = w 2σ 2 + w 2σ 2 + w w σ 2 = (½)2(100) + (½)2(100) + (½)(½)(50) = ¼(100 + 100 P 1 1 2 2 1 2 1,2 + 50) = 250/4 = 62.5 [%2], which gives a portfolio standard deviation of σ(R) = √(0.00625) P = 0.07906 ≈ 0.079 = 7.9%.

B. 8.7%. - Correct

Correct Answer Feedback: Correct because σ2(R) = Σ n Σ n wwCov(R,R). P i=1 j=1 i j i j For two assets, n = 2, σ2(R) = w 2σ 2 + w 2σ 2 + 2w w σ 2, P 1 1 2 2 1 2 1,2 where, from given values, σ 2 = σ 2 = 100%2, σ 2 = 50%2. 1 2 1,2 Since the portfolio is equally weighted, w = 1/n = ½. i Hence, the portfolio variance, σ2(R) = (½)2(100)

Incorrect Answer Feedback: Incorrect because it is the portfolio standard deviation calculated from an incorrect formula for portfolio variance, where the "2" is omitted from the covariance term, i.e.

$\sigma^2(R_P) = w_1{}^2\sigma_1{}^2 + w_2{}^2\sigma_2{}^2 + w_1w_2\sigma_{1,2}{}^2 = (½)^2(100) + (½)^2(100) + (½)(½)(50) = ¼(100 + 100 + 50) = 250/4 = 62.5\ [\%^2]$, which gives a portfolio standard deviation of $\sigma(R_P) = \sqrt{(0.00625)} = 0.07906 \approx 0.079 = 7.9\%$.

Figure 29: Source equation for question 57, option A

+ (½)2(100) + 2(½)(½)(50) = ¼(100 + 100 P + 100) = 300/4 = 75 [%2], i.e. portfolio standard deviation, σ(R) = √(0.0075) = 0.08660 ≈ P 0.087 = 8.7%.

Correct Answer Feedback: Correct because $\sigma^2(R_P) = \Sigma_{i=1}{}^n \Sigma_{j=1}{}^n w_iw_j\text{Cov}(R_i,R_j)$.

For two assets, $n = 2$, $\sigma^2(R_P) = w_1{}^2\sigma_1{}^2 + w_2{}^2\sigma_2{}^2 + 2w_1w_2\sigma_{1,2}{}^2$,

where, from given values, $\sigma_1{}^2 = \sigma_2{}^2 = 100\%^2$, $\sigma_{1,2}{}^2 = 50\%^2$.

Since the portfolio is equally weighted, $w_i = 1/n = ½$.

Hence, the portfolio variance, $\sigma^2(R_P) = (½)^2(100) + (½)^2(100) + 2(½)(½)(50) = ¼(100 + 100 + 100) = 300/4 = 75\ [\%^2]$, i.e. portfolio standard deviation, $\sigma(R_P) = \sqrt{(0.0075)} = 0.08660 \approx 0.087 = 8.7\%$.

Figure 30: Source equation for question 57, option B

C. 75.0%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the portfolio variance σ2(R) = P (½)2(100) + (½)2(100) + 2(½)(½)(50) = 75 [%2], incorrectly expressed in %, and not the portfolio standard deviation.

Incorrect Answer Feedback: Incorrect because it is the portfolio variance $\sigma^2(R_P) = (½)^2(100) + (½)^2(100) + 2(½)(½)(50) = 75\ [\%^2]$, incorrectly expressed in %, and not the portfolio standard deviation.

(no answer)

Figure 31: Source equation for question 57, option C

Question 58

In capital investment decisions, an abandonment option is most likely a type of:

Correct answer: A

A. sizing option. - Correct

Correct Answer Feedback: Correct because an abandonment option is a type of sizing option. If after investing the company can abandon the investment if the financial results are disappointing, it has an abandonment option.

B. timing option. - Incorrect

Incorrect Answer Feedback: Incorrect because an abandonment option is a type of sizing option, not a type of timing option. A timing option means that instead of investing now, the company can delay investing. Delaying an investment and basing the decision on hopefully improved information could help improve the NPV of the projects selected.

C. flexibility option. - Incorrect

Incorrect Answer Feedback: Incorrect because an abandonment option is a type of sizing option, not a type of flexibility option. Once an investment is made, operational flexibilities besides abandonment or expansion may be available. Management may be able to exercise a price-setting option, which is a flexibility option to benefit from excess demand. There are also production flexibility options which offer the operational flexibility to alter production when demand varies from what is forecast.

Question 59

Which of the following available sources of capital is taken into account to calculate the WACC of an unlevered company?

Correct answer: C

A. Preferred stock only - Incorrect

Incorrect Answer Feedback: Incorrect because WACC = (Cost of debt × Weighting of debt) + (Cost of equity × Weighting of equity). The cost of equity is included in the calculation.

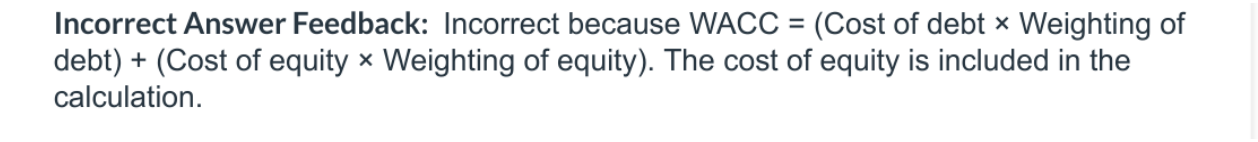


Figure 32: Source equation for question 59, option A

B. Common stock only - Incorrect

Incorrect Answer Feedback: Incorrect because WACC = (Cost of debt × Weighting of debt) + (Cost of equity × Weighting of equity). If an issuer has additional types of financing with different risk and return characteristics from those of debt and equity—for example, preferred stock and non-controlling interests—the weighted-average after-tax cost of that financing needs to be added to the equation to calculate WACC. Therefore, the cost of preferred must also be included in the calculation.

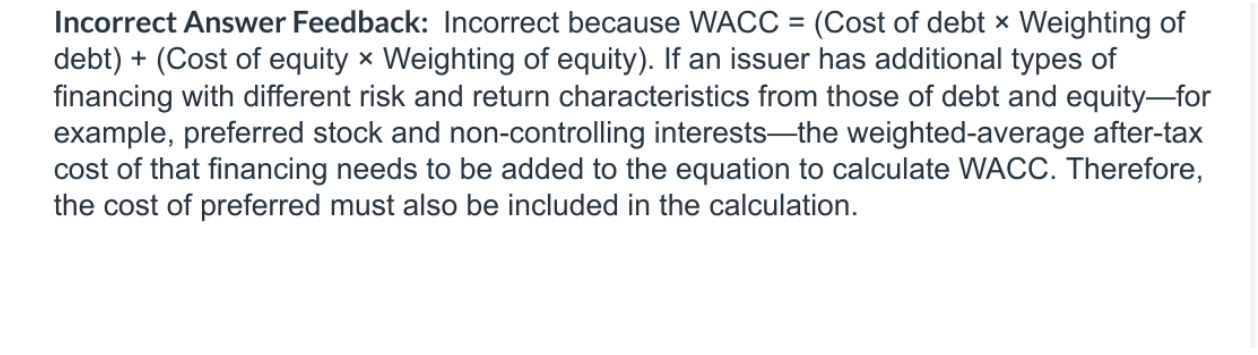


Figure 33: Source equation for question 59, option B



Figure 34: Source equation for question 59, option B

C. Both preferred stock and common stock - Correct

Correct Answer Feedback: Correct because WACC = (Cost of debt × Weighting of debt) + (Cost of equity × Weighting of equity). If an issuer has additional types of financing with different risk and return characteristics from those of debt and equity— for example, preferred stock and non-controlling interests—the weighted-average after-tax cost of that financing needs to be added to the equation to calculate WACC. Therefore, the cost of preferred and the cost of equity are included in the calculation.

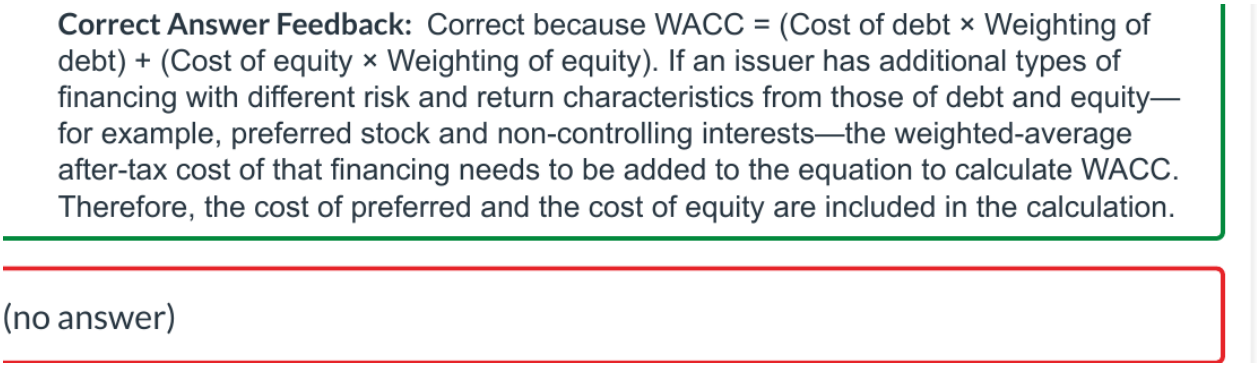


Figure 35: Source equation for question 59, option C

Question 60

An analyst gathers the following information (in € thousands) about an electronics manufacturing company:

	Year 2	Year 1
Cost of sales	1,250	1,000
Cost of inventory	225	375
Net realizable value of inventory	300	325

Changes to the allowance for inventory obsolescence have already been reflected in cost of sales. The inventory turnover (based on average inventory) for Year 2 is closest to:

Correct answer: C

A. 4.0. - Incorrect

Incorrect Answer Feedback: Incorrect because inventories in both years were measured at net realizable value. Hence, Inventory turnover ratio = 1,250 / [(300 + 325) / 2] = 1,250 / 312.5 = 4.0.

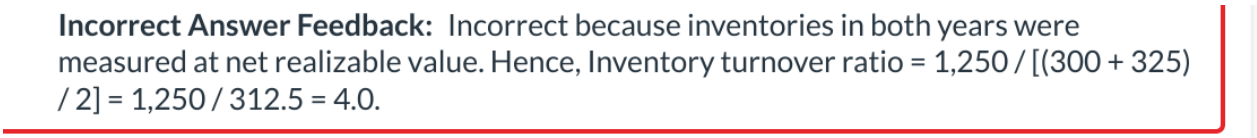


Figure 36: Source equation for question 60, option A

B. 4.2. - Incorrect

Incorrect Answer Feedback: Incorrect because inventories in both years were measured at cost. Hence, Inventory turnover ratio = 1,250 / [(225 + 375) / 2] = 1,250 / 300 = 4.167 ≈ 4.2.

C. 4.5. - Correct

Correct Answer Feedback: Correct because IFRS state that inventories shall be measured (and carried on the balance sheet) at the lower cost and net realizable value and Inventory turnover ratio = Cost of sales ÷ Average inventory. Accordingly, Inventory turnover ratio = 1,250 / [(225 + 325)] / 2 = 1,250 / 275 = 4.545 ≈ 4.5.

Incorrect Answer Feedback: Incorrect because inventories in both years were measured at cost. Hence, Inventory turnover ratio = 1,250 / [(225 + 375) / 2] = 1,250 / 300 = 4.167 ≈ 4.2.

Figure 37: Source equation for question 60, option B

Correct Answer Feedback: Correct because IFRS state that inventories shall be measured (and carried on the balance sheet) at the lower cost and net realizable value and Inventory turnover ratio = Cost of sales ÷ Average inventory. Accordingly, Inventory turnover ratio = 1,250 / [(225 + 325)] / 2 = 1,250 / 275 = 4.545 ≈ 4.5.

Figure 38: Source equation for question 60, option C

Question 61

The long-run marginal cost schedule serves as a firm's supply curve under a(n):

Correct answer: B

A. oligopoly market. - Incorrect

Incorrect Answer Feedback: Incorrect because as in monopolistic competition, the oligopolist does not have a well-defined supply function. That is, there is no way to determine the oligopolist's optimal levels of output and price independent of demand conditions and competitor's strategies.

B. perfectly competitive market. - Correct

Correct Answer Feedback: Correct because the long-run marginal cost schedule is the perfectly competitive firm's supply curve.

C. monopolistically competitive market. - Incorrect

Incorrect Answer Feedback: Incorrect because in monopolistic competition, there is no well-defined supply function. The firm's supply curve should measure the quantity the firm is willing to supply at various prices. That information is not represented by either marginal cost or average cost.

Question 62

Fiat money is currency that is:

Correct answer: A

A. legal tender. - Correct

Correct Answer Feedback: Correct because money in all major economies today is not convertible by law into anything else, but it is, in law, legal tender. This means that it must be accepted when offered in exchange for goods and services. Money that is not convertible into any other commodity is known as fiat money.

B. not created by a central bank. - Incorrect

Incorrect Answer Feedback: Incorrect because a central bank is the monopoly supplier of the currency. Although a bank creates money when it makes a loan, currency is created when the bank exchanges reserves for currency at the central bank.

C. convertible by law into gold or any other commodity. - Incorrect

Incorrect Answer Feedback: Incorrect because money that is not convertible into any other commodity is known as fiat money.

Question 63

The reversal of an inventory write-down:

Correct answer: A

A. reduces cost of sales. - Correct

Correct Answer Feedback: Correct because in each period subsequent to inventory write-down, a new assessment of net realisable value is made. Reversal (limited to the amount of original write-down) is required for a subsequent increase in value of inventory previously written down. The reversal of any write-down of inventories is recognized as a reduction in cost of sales (reduction in the amount of inventories recognized as an expense).

B. increases other comprehensive income. - Incorrect

Incorrect Answer Feedback: Incorrect because five items are treated as other comprehensive income under IFRS: foreign currency translation adjustments in consolidating financial statements of foreign subsidiaries, unrealized gains or losses on derivative contracts accounted for as hedges, unrealized gains and losses on available for sale securities and certain costs of a company's defined benefit post-retirement plans that are not recognized in the current period and gains or losses from revaluation of fixed assets. The reversal of inventory write-downs is not one of them. The reversal of any write-down of inventories is recognized as a reduction in cost of sales (reduction in the amount of inventories recognized as an expense).

C. is permitted under both IFRS and US GAAP. - Incorrect

Incorrect Answer Feedback: Incorrect because under IFRS, if inventory that was written down in a previous period subsequently increases in value, the amount of the original writedown is reversed. Subsequent reversal of an inventory write-down is not permitted under US GAAP.

Question 64

An asset earns the following monthly returns: January −30% February 10% March 10% April 10% The sample skewness is:

Correct answer: A

A. negative. - Correct

Correct Answer Feedback: Correct because the sample skewness is negative as a return distribution with negative skew has frequent small gains and a few extreme losses. Negative skewness can also be inferred because the arithmetic mean return, i.e., , is less than the median return (10%). For the continuous negatively skewed unimodal distribution, the mean is less than the median. Alternatively, the formula for the skewness coefficient could be used; . It suffices to recognize that the sum of the cubed deviations from the mean (the numerator) is negative; $(-0.3)3 + 0.13 + 0.13 + 0.13 = -0.024$. Therefore, the sample distribution is negatively skewed.

B. zero. - Incorrect

Incorrect Answer Feedback: Incorrect because the sample skewness is negative, not zero. A candidate may incorrectly conclude zero skewness because the sum of the positive returns is equal to the sum of the negative returns.

C. positive. - Incorrect

Incorrect Answer Feedback: Incorrect because the sample skewness is negative, not positive. A candidate may incorrectly conclude positive skewness because the sample has more positive returns than negative returns.

Correct Answer Feedback: Correct because the sample skewness is negative as a return distribution with negative skew has frequent small gains and a few extreme losses. Negative skewness can also be inferred because the arithmetic mean return, i.e., $\frac{(-30\%)+10\%+10\%+10\%}{4} = 0\%$, is less than the median return (10%). For the continuous negatively skewed unimodal distribution, the mean is less than the median. Alternatively, the formula for the skewness coefficient could be used; $S_k = \left(\frac{n}{(n-1)(n-2)}\right)\frac{\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^3}{s^3}$. It suffices to recognize that the sum of the cubed deviations from the mean (the numerator) is negative; $(-0.3)^3 + 0.1^3 + 0.1^3 + 0.1^3 = -0.024$. Therefore, the sample distribution is negatively skewed.

Figure 39: Source equation for question 64, option A

Question 65

Which of the following statements about the use of financial ratios is most accurate?

Correct answer: B

A. Each financial ratio is relevant to all industries - Incorrect

Incorrect Answer Feedback: Incorrect because many ratios are industry specific, and not all ratios are important to all industries.

B. Financial ratios can be affected by the phase of the business cycle - Correct

Correct Answer Feedback: Correct because both financial ratios (for cyclical companies) tend to improve when the economy is strong and weaken during recessions. Therefore, financial ratios should be examined in light of the current phase of the business cycle.

C. Aggregate financial ratios are useful for companies operating in different lines of business - Incorrect

Incorrect Answer Feedback: Incorrect because companies may have several different lines of business. This will cause aggregate financial ratios to be distorted. It is better to examine industry-specific ratios by lines of business.

Question 66

A decrease in which of the following ratios decreases the cash conversion cycle?

Correct answer: A

A. Payables turnover - Correct

Correct Answer Feedback: Correct because the cash conversion cycle = days of inventory on hand (DOH) + days of sales outstanding (DSO) – number of days of payables. Since DOH = (Number of days in period) / (Inventory turnover), DSO = (Number of days in period) / (Receivables turnover), and number of days of payables = (Number of days in period) / (Payables turnover), the cash conversion cycle = (Number of days in period) / (Inventory turnover) + (Number of days in period) / (Receivables turnover) - (Number of days in period) / (Payables turnover). If payables turnover decreases (i.e., the denominator of the number of days of payables formula), the number of days of payables will increase since the numerator (number of days in period) stays the same. Because the number of days of payables is the only subtraction item in the cash conversion cycle formula, an increase in in this number will shorten the cash conversion cycle.

B. Inventory turnover - Incorrect

Incorrect Answer Feedback: Incorrect because a decrease in inventory turnover would increase DOH (an addition in the cash conversion cycle formula), which would increase the cash conversion cycle.

Correct Answer Feedback: Correct because the cash conversion cycle = days of inventory on hand (DOH) + days of sales outstanding (DSO) - number of days of payables.

Since DOH = (Number of days in period) / (Inventory turnover),

DSO = (Number of days in period) / (Receivables turnover),

and number of days of payables = (Number of days in period) / (Payables turnover),

the cash conversion cycle = (Number of days in period) / (Inventory turnover) + (Number of days in period) / (Receivables turnover) - (Number of days in period) / (Payables turnover).

If payables turnover decreases (i.e., the denominator of the number of days of payables formula), the number of days of payables will increase since the numerator (number of days in period) stays the same. Because the number of days of payables is the only subtraction item in the cash conversion cycle formula, an increase in in this number will shorten the cash conversion cycle.

Figure 40: Source equation for question 66, option A

C. Receivables turnover - Incorrect

Incorrect Answer Feedback: Incorrect because a decrease in receivables turnover would increase DSO (an addition in the cash conversion cycle formula), which would increase the cash conversion cycle.

Question 67

Frequent changes to commercial bank reserve requirements most likely:

Correct answer: A

A. are disruptive for banks. - Correct

Correct Answer Feedback: Correct because changing reserve requirements frequently is disruptive for banks. For example, if a central bank increased the reserve requirements, a bank that was short on reserves might have to cease its lending activities until it had built up the necessary reserves, because deposits would be unlikely to rise quickly enough for the bank to build its reserves in this way.

B. lead to high money growth. - Incorrect

Incorrect Answer Feedback: Incorrect because although the money creation process is more powerful the lower the percentage reserve requirement of banks, this is not directly related to the frequency of changes and reserve requirement changes can potentially lead to higher or lower money growth.

C. are common in developed countries. - Incorrect

Incorrect Answer Feedback: Incorrect because this policy tool (reserve requirement of banks) is not used much nowadays in developed economies. However, reserve requirements are still actively used in many developing countries to control lending and remain a potential policy tool for those central banks that do not currently use it.

Question 68

In the long run, a perfectly competitive firm's marginal cost is:

Correct answer: B

A. less than its average cost. - Incorrect

Incorrect Answer Feedback: Incorrect because at the level of output at which SMC (short-run marginal cost) is equal to MR (marginal revenue), price is equal to (not less than) ATC (average total cost).

B. equal to its average cost. - Correct

Correct Answer Feedback: Correct because in the long run in a perfectly competitive market, competitors would enter the market to capture any profits and would drive the market price down to the level equal to each firm's ATC (average total cost). At the level of output at which SMC (short-run marginal cost) is equal to MR (marginal revenue), price is equal to ATC (average total cost).

C. greater than its average cost. - Incorrect

Incorrect Answer Feedback: Incorrect because at the level of output at which SMC (short-run marginal cost) is equal to MR (marginal revenue), price is equal to (not greater than) ATC (average total cost).

Question 69

With respect to Big Data, which of the following is best classified as structured data?

Correct answer: A

A. SQL tables - Correct

Correct Answer Feedback: Correct because the term Big Data has been in use since the late 1990s and refers to the vast amount of data being generated by industry, governments, individuals, and electronic devices. The data are collected from many different sources and in a variety of formats, including structured data (e.g., SQL tables or CSV files), semi-structured data (e.g., HTML code), and unstructured data (e.g., video messages). Structured data items can be organized in tables and are commonly stored in a database where each field represents the same type of information.

B. HTML code - Incorrect

Incorrect Answer Feedback: Incorrect because data are collected from many different sources and in a variety of formats, including structured data (e.g., SQL tables or CSV files), semi-structured data (e.g., HTML code), and unstructured data (e.g., video messages). Semi-structured data may have attributes of both structured and unstructured data.

C. Video messages - Incorrect

Incorrect Answer Feedback: Incorrect because data are collected from many different sources and in a variety of formats, including structured data (e.g., SQL tables or CSV files), semi-structured data (e.g., HTML code), and unstructured data (e.g., video messages). Unstructured data may be disparate, unorganized data that cannot be represented in tabular form. Unstructured data, such as those generated by social media, email, text messages, voice recordings, pictures, blogs, scanners, and sensors, often require different, specialized applications or custom programs before they can be useful to investment professionals.

Question 70

An analyst gathers the following information (in € thousands) about a company's equipment:

Carrying amount prior to impairment	20,000
Undiscounted expected future cash flows	19,000
Present value of expected future cash flows	17,000
Fair value if sold	18,000
Costs to sell	2,000

Based only on this information, the impairment loss under IFRS is:

Correct answer: C

A. less than the impairment loss under US GAAP. - Incorrect

Incorrect Answer Feedback: Incorrect because impairment loss is higher under IFRS than GAAP. Under IFRS, the company would compare the carrying amount (€20,000) with the higher of its fair value less costs to sell (€16,000) and its value in use (€17,000). The carrying amount exceeds the value in use, the higher of the two amounts, by $3,000. The machine would be written down to the recoverable amount of €17,000, and an impairment loss of €3,000 would be reported in the income statement. The carrying amount of the machine is now €17,000. Under US GAAP, the carrying amount (€20,000) is compared with the undiscounted expected future cash flows (€19,000). The carrying amount exceeds the undiscounted expected future cash flows, so the carrying amount is considered not recoverable. The machine would be written down to fair value of €18,000 and an impairment loss of €2,000 would be reported in the income statement. The carrying amount of the machine is now €18,000.

B. the same as the impairment loss under US GAAP. - Incorrect

Incorrect Answer Feedback: Incorrect because impairment loss is higher under IFRS than GAAP. Under IFRS, the company would compare the carrying amount (€20,000) with the higher of its fair value less costs to sell (€16,000) and its value in use (€17,000). The carrying amount exceeds the value in use, the higher of the two amounts, by €3,000. The machine would be written down to the recoverable amount of €17,000, and an impairment loss of €3,000 would be reported in the income statement. The carrying amount of the machine is now €17,000. Under US GAAP, the carrying amount (€20,000) is compared with the undiscounted expected future cash flows (€19,000). The carrying amount exceeds the undiscounted expected future cash flows, so the carrying amount is considered not recoverable. The machine would be written down to fair value of €18,000 and an impairment loss of €2,000 would be reported in the income statement. The carrying amount of the machine is now €18,000.

C. greater than the impairment loss under US GAAP. - Correct

Correct Answer Feedback: Correct because impairment loss is higher under IFRS than GAAP. Under IFRS, the company would compare the carrying amount (€20,000) with the greater of its fair value less costs to sell (€16,000) and its value in use (€17,000). The carrying amount exceeds the value in use, the higher of the two amounts, by €3,000. The machine would be written down to the recoverable amount of €17,000, and an impairment loss of €3,000 would be reported in the income statement. The carrying amount of the machine is now €17,000. Under US GAAP, the carrying amount (€20,000) is compared with the undiscounted expected future cash flows (€19,000). The carrying amount exceeds the undiscounted expected future cash flows, so the carrying amount is considered not recoverable. The machine would be written down to fair value of €18,000 and an impairment loss of €2,000 would be reported in the income statement. The carrying amount of the machine is now €18,000.

Question 71

When revenue is recognized for a contract with separate performance obligations that have not all been met, and no consideration has been received, the seller’s balance sheet will reflect a:

Correct answer: B

A. receivable. - Incorrect

Incorrect Answer Feedback: Incorrect because it is only at the point when all performance obligations have been met except for payment that a receivable appears on the seller’s balance sheet.

B. contract asset. - Correct

Correct Answer Feedback: Correct because revenue is recognized when a performance obligation is fulfilled. When revenue is recognized, a contract asset is presented on the balance sheet. It is only at the point when all performance obligations have been met except for payment that a receivable appears on the seller’s balance sheet.

C. contract liability. - Incorrect

Incorrect Answer Feedback: Incorrect because when revenue is recognized, a contract asset is presented on the balance sheet. Also, if consideration is received in advance of transferring good(s) or service(s), the seller presents a contract liability.

Question 72

An analyst gathers the following return data: 1.4% 2.7% 4.1% 4.8% 6.1% 6.9% 8.3% 9.0% th The 30 percentile is closest to:

Correct answer: C

A. 2.7%. - Incorrect

Incorrect Answer Feedback: Incorrect because it assumes the 30th percentile is 30% of the largest observation; 0.30 × (9.0%) = 2.7%. This is also the location of the 30th percentile and the value of the second observation.

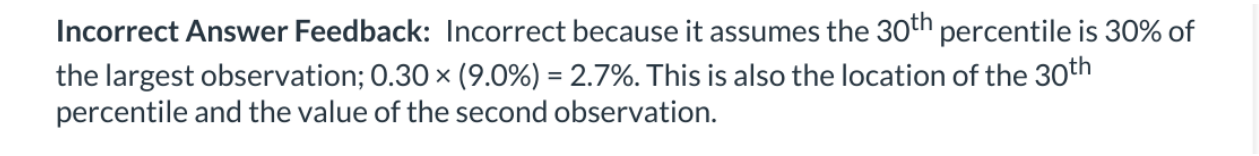


Figure 41: Source equation for question 72, option A

B. 3.3%. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses the sample size instead of n + 1 in the percentile formula; 8 × (30 / 100) = 2.4; 2.7% + [(2.4 – 2.0) × (4.1% − 2.7%)] = 3.3%.

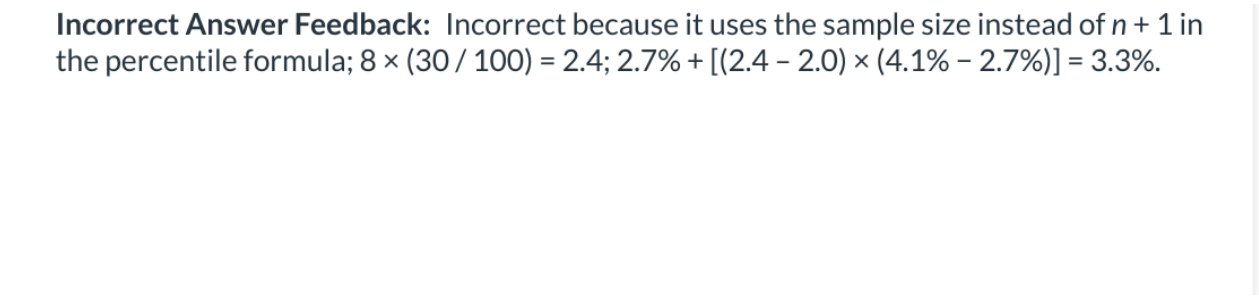


Figure 42: Source equation for question 72, option B



Figure 43: Source equation for question 72, option B

C. 3.7%. - Correct

Correct Answer Feedback: Correct because the yth percentile is the value at or below which y% of observations lie. The location of a percentile is: (n + 1) × (y / 100); here, the location of the 30th percentile is (8 + 1) × (30 / 100) = 2.7. The value of 30th percentile is determined using linear interpolation: 2.7% + [(2.7 – 2.0) × (4.1% – 2.7%)] = 3.7%.

Question 73

In a simple linear regression, the standard error of the estimate is also known as the:

Correct answer: C

A. mean square error. - Incorrect

Incorrect Answer Feedback: Incorrect because the standard error of the estimate (s), is e also known as the standard error of the regression or the root mean square error. In fact, the standard error of the estimate is equal to the square root of the mean square error. However, the standard error of the estimate is not another name for the mean square error.

Correct Answer Feedback: Correct because the *y*th percentile is the value at or below which y% of observations lie. The location of a percentile is: $(n + 1) \times (y / 100)$; here, the location of the 30th percentile is $(8 + 1) \times (30 / 100) = 2.7$. The value of 30th percentile is determined using linear interpolation: $2.7\% + [(2.7 - 2.0) \times (4.1\% - 2.7\%)] = 3.7\%$.

(no answer)

Figure 44: Source equation for question 72, option C

B. sum of squares error. - Incorrect

Incorrect Answer Feedback: Incorrect because the standard error of the estimate (s), is e also known as the standard error of the regression or the root mean square error. In fact, the standard error of the estimate is equal to the square root of the mean square error (MSE), which is the sum of squares error divided by the degrees of freedom. In other words, while the standard error of the estimate is related to the sum of squared error, it is not another name for it.

C. root mean square error. - Correct

Correct Answer Feedback: Correct because the standard error of the estimate (s), is e also known as the standard error of the regression or the root mean square error.

Question 74

According to the Modigliani–Miller propositions, if there are no taxes, an increase in financial leverage will increase the cost of equity and:

Correct answer: B

A. decrease the WACC. - Incorrect

Incorrect Answer Feedback: Incorrect because in the Modigliani-Miller propositions, an increase in financial leverage will increase the cost of equity, but will not change the WACC in the absence of taxes. The market value of a company is not affected by the company's capital structure. In the absence of taxes, the weighted average cost of capital (WACC), or WACC = w r + w r + w r , is unaffected by capital structure. However, in the d d p p e e presence of taxes, an increase in financial leverage will decrease the cost of capital. Consequently, as debt increases, the company's WACC decreases and the company's value increases.

Incorrect Answer Feedback: Incorrect because in the Modigliani-Miller propositions, an increase in financial leverage will increase the cost of equity, but will not change the WACC in the absence of taxes. The market value of a company is not affected by the company's capital structure. In the absence of taxes, the weighted average cost of capital (WACC), or WACC = $w_d r_d + w_p r_p + w_e r_e$, is unaffected by capital structure. However, in the presence of taxes, an increase in financial leverage will decrease the cost of capital. Consequently, as debt increases, the company's WACC decreases and the company's value increases.

Figure 45: Source equation for question 74, option A

B. have no effect on the WACC. - Correct

Correct Answer Feedback: Correct because in the Modigliani-Miller propositions, an increase in financial leverage will increase the cost of equity, but will not change the WACC in the absence of taxes. The market value of a company is not affected by the company's capital structure. In the absence of taxes, the weighted average cost of capital (WACC), or WACC = w r + w r + w r , is unaffected by capital structure. d d p p e e

Correct Answer Feedback: Correct because in the Modigliani-Miller propositions, an increase in financial leverage will increase the cost of equity, but will not change the WACC in the absence of taxes. The market value of a company is not affected by the company's capital structure. In the absence of taxes, the weighted average cost of capital (WACC), or WACC = $w_d r_d + w_p r_p + w_e r_e$, is unaffected by capital structure.

Figure 46: Source equation for question 74, option B

C. increase the WACC. - Incorrect

Incorrect Answer Feedback: Incorrect because in the Modigliani-Miller propositions, an increase in financial leverage will increase the cost of equity, but will not change the WACC in the absence of taxes. The market value of a company is not affected by the company's capital structure. In the absence of taxes, the weighted average cost of capital (WACC), or WACC = w r + w r + w r , is unaffected by capital structure. d d p p e e

Incorrect Answer Feedback: Incorrect because in the Modigliani-Miller propositions, an increase in financial leverage will increase the cost of equity, but will not change the WACC in the absence of taxes. The market value of a company is not affected by the company's capital structure. In the absence of taxes, the weighted average cost of capital (WACC), or WACC = $w_d r_d + w_p r_p + w_e r_e$, is unaffected by capital structure.

(no answer)

Figure 47: Source equation for question 74, option C

Question 75

If a deferred tax asset resulted in the past, but the criteria of economic benefits is not met on the current balance sheet date, the company should:

Correct answer: B

A. perform an impairment test. - Incorrect

Incorrect Answer Feedback: Incorrect because if a deferred tax asset or liability resulted in the past, but the criteria of economic benefits is not met on the current balance sheet date, then, under IFRS, an existing deferred tax asset or liability related to the item will be reversed.

B. reverse the deferred tax asset. - Correct

Correct Answer Feedback: Correct because if a deferred tax asset or liability resulted in the past, but the criteria of economic benefits is not met on the current balance sheet date, then, under IFRS, an existing deferred tax asset or liability related to the item will be reversed.

C. establish a valuation allowance. - Incorrect

Incorrect Answer Feedback: Incorrect because if a deferred tax asset or liability resulted in the past, but the criteria of economic benefits is not met on the current balance sheet date, then, under IFRS, an existing deferred tax asset or liability related to the item will be reversed. Under US GAAP, a valuation allowance is established.

Question 76

In an economy with long-run expected inflation of 3% and a real trend rate of growth of 2%, the neutral policy rate is closest to:

Correct answer: C

A. 1%. - Incorrect

Incorrect Answer Feedback: Incorrect because it subtracts the real trend rate of growth from expected inflation to arrive at the neutral rate (rather than adding it); 3% − 2% = 1%.

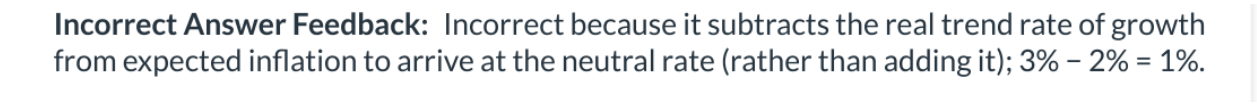


Figure 48: Source equation for question 76, option A

B. 3%. - Incorrect

Incorrect Answer Feedback: Incorrect because it assumes the neutral rate is equal to expected inflation (3%) and does not include the real trend rate of growth.

C. 5%. - Correct

Correct Answer Feedback: Correct because the neutral policy rate for any economy comprises two components: Real trend rate of growth of the underlying economy, and long-run expected inflation. Therefore, the correct calculation is: Neutral rate = Trend growth + Inflation target = 2% + 3% = 5%.

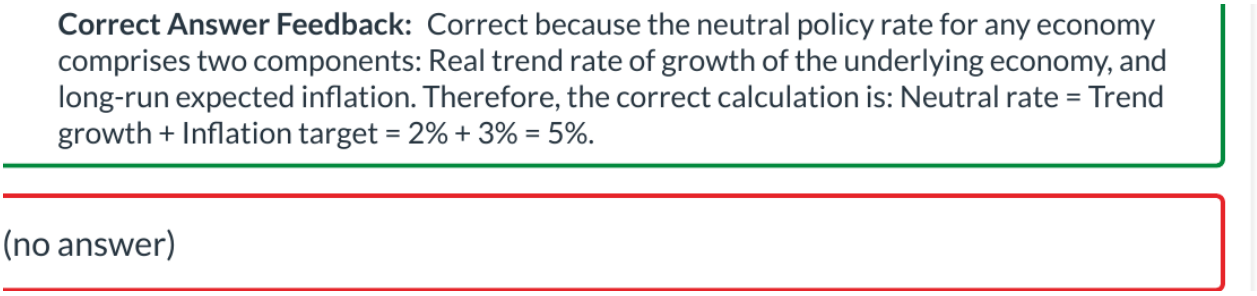


Figure 49: Source equation for question 76, option C

Question 77

Which is the first step in the standard process of hypothesis testing?

Correct answer: B

A. Collect data - Incorrect

Incorrect Answer Feedback: Incorrect because, according to the standard approach to hypothesis testing uses the following process, the first step is “State the hypotheses”. Collecting data is Step 5.

B. State the hypothesis - Correct

Correct Answer Feedback: Correct because the process of hypothesis testing begins with the formulation of a theory to organize and explain observations. The standard approach to hypothesis testing uses the following process: Step 1: State the hypotheses; Step 2: Identify the appropriate test statistic; Step 3: Specify the level of significance; Step 4: State the decision rule; Step 5: Collect data and calculate the test statistic; Step 6: Make a decision. Thus, in the standard approach to hypothesis testing, the first step is to state the hypothesis.

C. Specify the level of significance - Incorrect

Incorrect Answer Feedback: Incorrect because, according to the standard approach to hypothesis testing uses the following process, the first step is “State the hypotheses”. Specifying the level of significance is Step 3.

Question 78

Which of the following events is most likely to lead management to make biased accounting choices?

Correct answer: A

A. Changing the company's CEO - Correct

Correct Answer Feedback: Correct because a company might report lower earnings in the CEO's first year to create a positive trajectory for future periods. Alternatively, there could be motivation to report higher earnings under the new CEO. Either situation could lead to biased accounting choices.

B. Changing the company's audit firm - Incorrect

Incorrect Answer Feedback: Incorrect because changing the audit firm is not likely to lead to biased reporting.

C. Changing the company's fiscal year - Incorrect

Incorrect Answer Feedback: Incorrect because changing the fiscal year is not likely to lead to biased reporting.

Question 79

A lessor would derecognize the underlying leased asset from its balance sheet and recognize a lease receivable for:

Correct answer: A

A. a finance lease only. - Correct

Correct Answer Feedback: Correct because the accounting for lessors is identical under IFRS and US GAAP. Under both accounting standards, lessors classify leases as finance or operating leases, which determines the financial reporting. At finance lease inception, the lessor recognizes a lease receivable asset equal to the present value of future lease payments and de-recognizes the leased asset, simultaneously recognizing any difference as a gain or loss.

B. an operating lease only. - Incorrect

Incorrect Answer Feedback: Incorrect because the accounting for lessors is identical under IFRS and US GAAP. When accounting for an operating lease, because the contract is essentially a rental agreement, the lessor keeps the leased asset on its books and recognizes lease revenue on a straight-line basis.

C. both a finance lease and an operating lease. - Incorrect

Incorrect Answer Feedback: Incorrect because the accounting for lessors is identical under IFRS and US GAAP. Under both accounting standards, lessors classify leases as finance or operating leases, which determines the financial reporting. At finance lease inception, the lessor recognizes a lease receivable asset equal to the present value of future lease payments and de-recognizes the leased asset, simultaneously recognizing any difference as a gain or loss. The accounting treatment for an operating lease is different in that, because the contract is essentially a rental agreement, the lessor keeps the leased asset on its books and recognizes lease revenue on a straight- line basis.

Question 80

All else being equal, in periods of rising prices and constant inventory quantities, FIFO most likely results in a higher:

Correct answer: C

A. total asset turnover than LIFO. - Incorrect

Incorrect Answer Feedback: Incorrect because Total asset turnover = Revenue ÷ Average total assets. As inventory is higher under FIFO relative to LIFO, Average total assets is higher. Revenue is not affected by the choice of inventory valuation method. As such, Total asset turnover is lower under FIFO when compared to LIFO.

Incorrect Answer Feedback: Incorrect because Total asset turnover = Revenue ÷ Average total assets. As inventory is higher under FIFO relative to LIFO, Average total assets is higher. Revenue is not affected by the choice of inventory valuation method. As such, Total asset turnover is lower under FIFO when compared to LIFO.

Figure 50: Source equation for question 80, option A

B. working capital turnover than LIFO. - Incorrect

Incorrect Answer Feedback: Incorrect because Working capital turnover = Revenue ÷ Average working capital. As inventory is higher under FIFO relative to LIFO, Average working capital is higher. Revenue is not affected by the choice of inventory valuation method. As such, Working capital turnover is lower under FIFO when compared to LIFO.

Incorrect Answer Feedback: Incorrect because Working capital turnover = Revenue ÷ Average working capital. As inventory is higher under FIFO relative to LIFO, Average working capital is higher. Revenue is not affected by the choice of inventory valuation method. As such, Working capital turnover is lower under FIFO when compared to LIFO.

Figure 51: Source equation for question 80, option B

C. days of inventory on hand than LIFO. - Correct

Correct Answer Feedback: Correct because in an environment of rising inventory unit costs and constant or increasing inventory quantities, FIFO (in comparison with weighted average cost or LIFO) will allocate a lower amount of the total cost of goods available for sale to cost of sales on the income statement and a higher amount to ending inventory on the balance sheet. Accordingly, because cost of sales will be lower under FIFO, a company's gross profit, operating profit, and income before taxes will be higher. Days of inventory on hand = Number of days in period ÷ Inventory turnover. Inventory turnover = Cost of sales ÷ Average inventory. As inventory is higher under FIFO relative to LIFO, and Cost of sales is lower under FIFO relative to LIFO, Inventory turnover is lower under FIFO. As such, Days of inventory on hand is higher under FIFO relative to LIFO.

Correct Answer Feedback: Correct because in an environment of rising inventory unit costs and constant or increasing inventory quantities, FIFO (in comparison with weighted average cost or LIFO) will allocate a lower amount of the total cost of goods available for sale to cost of sales on the income statement and a higher amount to ending inventory on the balance sheet. Accordingly, because cost of sales will be lower under FIFO, a company's gross profit, operating profit, and income before taxes will be higher. Days of inventory on hand = Number of days in period ÷ Inventory turnover. Inventory turnover = Cost of sales ÷ Average inventory. As inventory is higher under FIFO relative to LIFO, and Cost of sales is lower under FIFO relative to LIFO, Inventory turnover is lower under FIFO. As such, Days of inventory on hand is higher under FIFO relative to LIFO.

(no answer)

Figure 52: Source equation for question 80, option C

Question 81

An analyst gathers the following information about a company:

	Year 2	Year 1
Net profit margin	3.6%	2.5%
Total asset turnover	1.0	1.5
Financial leverage	1.25	1.2

Based only on this information, from Year 1 to Year 2, the company's ROE:

Correct answer: B

A. decreased. - Incorrect

Incorrect Answer Feedback: Incorrect because ROE remained the same from Year 1 to Year 2 instead of decreased. One possible way to calculate a decrease is to calculate ROE = (Net profit margin × Total asset turnover) / Leverage. Consequently, Year 1 ROE = (2.5% × 1.5) / 1.2 = 3.125%, and Year 2 ROE = (3.6% × 1.0) / 1.25 = 2.88%.

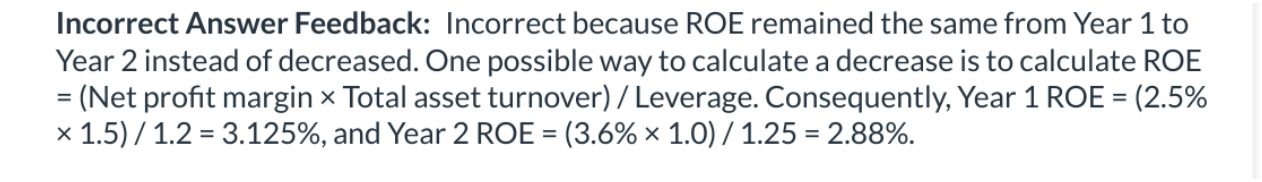


Figure 53: Source equation for question 81, option A

B. remained the same. - Correct

Correct Answer Feedback: Correct because "ROE = Net profit margin × Total asset turnover × Leverage". Accordingly, Year 1 ROE = 2.5% × 1.5 × 1.2 = 4.5%, and Year 2 ROE = 3.6% × 1.0 × 1.25 = 4.5%. Therefore, ROE remained the same from Year 1 to Year 2.

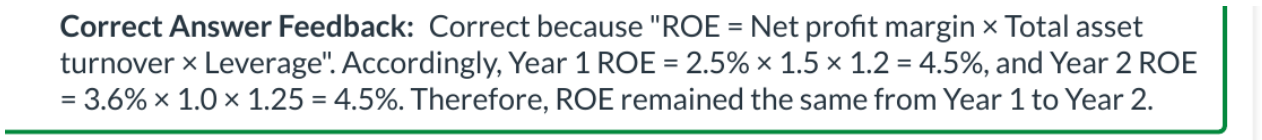


Figure 54: Source equation for question 81, option B

C. increased. - Incorrect

Incorrect Answer Feedback: Incorrect because ROE remained the same from Year 1 to Year 2 instead of increased. One possible way to calculate an increase is to calculate ROE = (Net profit margin × Leverage) / Total asset turnover. Consequently, Year 1 ROE = (2.5% × 1.2) / 1.5 = 2.0%, and Year 2 ROE = (3.6% × 1.25) / 1.0 = 4.5%.

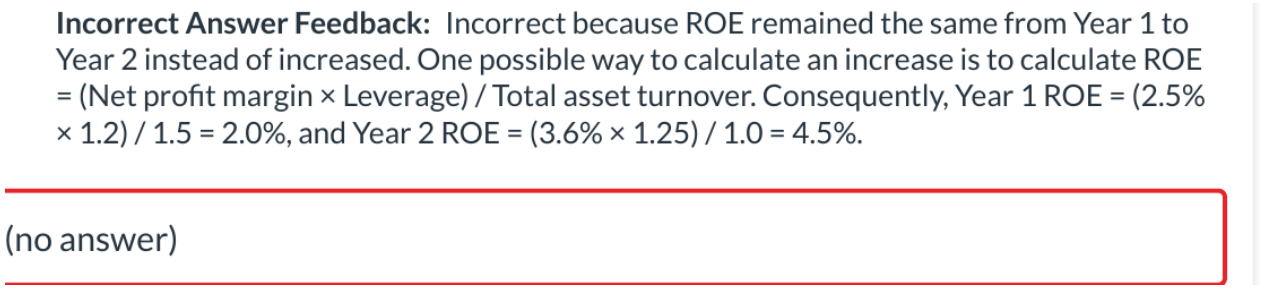


Figure 55: Source equation for question 81, option C

Question 82

Which of the following is a secondary source of liquidity?

Correct answer: B

A. Free cash flow - Incorrect

Incorrect Answer Feedback: Incorrect because primary sources of liquidity represent the most readily accessible resources available to the company. These can be cash held or near-cash securities and include free cash flow, which is the firm's after-tax operating cash flow less planned short- and long-term investments.

B. Dividend suspension - Correct

Correct Answer Feedback: Correct because the main difference between primary and secondary sources of liquidity is that using a primary source is not likely to affect the normal operations of the company, whereas using a secondary source might result in a change in the company's financial and operating positions. Secondary sources used by companies include negotiating debt contracts, such as relieving pressures from high interest payments or principal repayments, waiving debt covenants or suspending dividends.

C. Short-term investment portfolio - Incorrect

Incorrect Answer Feedback: Incorrect because primary sources of liquidity represent the most readily accessible resources available to the company. These can be cash held or near-cash securities and include short-term funds, which can include items such as trade credit, bank lines of credit, and short-term investment portfolios of the firm.

Question 83

An analyst gathers the following information (in £ millions) about a company's fiscal year:

Cost of goods sold	6,000
Increase in inventory	3,000
Increase in accounts payable	2,500

Cash paid to suppliers (in £ millions) is:

Correct answer: A

A. 6,500. - Correct

Correct Answer Feedback: Correct because Cash paid to suppliers = Cost of goods sold + Increase in inventory – Increase in accounts payable = 6,000 + 3,000 – 2,500 = 6,500.

Correct Answer Feedback: Correct because Cash paid to suppliers = Cost of goods sold + Increase in inventory – Increase in accounts payable = 6,000 + 3,000 – 2,500 = 6,500.

Figure 56: Source equation for question 83, option A

B. 9,000. - Incorrect

Incorrect Answer Feedback: Incorrect because it excluded the Increase in accounts payable and calculated Purchases from suppliers instead of Cash paid to suppliers. The calculation thus becomes Cash paid to suppliers = Purchases from suppliers = Cost of goods sold + Increase in inventory = 6,000 + 3,000 = 9,000.

C. 11,500. - Incorrect

Incorrect Answer Feedback: Incorrect because it added, rather than subtracted the Increase in accounts payable. The calculation thus becomes Cash paid to suppliers = Cost of goods sold + Increase in inventory + Increase in accounts payable = 6,000 + 3,000 + 2,500 = 11,500.

Incorrect Answer Feedback: Incorrect because it excluded the Increase in accounts payable and calculated Purchases from suppliers instead of Cash paid to suppliers. The calculation thus becomes Cash paid to suppliers = Purchases from suppliers = Cost of goods sold + Increase in inventory = 6,000 + 3,000 = 9,000.

Figure 57: Source equation for question 83, option B

Incorrect Answer Feedback: Incorrect because it added, rather than subtracted the Increase in accounts payable. The calculation thus becomes Cash paid to suppliers = Cost of goods sold + Increase in inventory + Increase in accounts payable = 6,000 + 3,000 + 2,500 = 11,500.

(no answer)

Figure 58: Source equation for question 83, option C

Question 84

If practical, retrospective application is required for a change in:

Correct answer: A

A. accounting policies only. - Correct

Correct Answer Feedback: Correct because changes in accounting policies are reported through retrospective application unless it is impractical to do so. Changes in accounting estimates, however, are handled prospectively.

B. accounting estimates only. - Incorrect

Incorrect Answer Feedback: Incorrect because companies sometimes make changes in accounting estimates (such as the useful life of a depreciable asset). Changes in accounting estimates are handled prospectively, with the change affecting the financial statements for the period of change and future periods. Changes in accounting policies, however, must be reported retrospectively unless it is impractical to do so.

C. both accounting policies and accounting estimates. - Incorrect

Incorrect Answer Feedback: Incorrect because changes in accounting estimates are handled prospectively, with the change affecting the financial statements for the period of change and future periods. Changes in accounting policies, however, must be reported retrospectively unless it is impractical to do so.

Question 85

An investment makes ten payments of $10,000 per year, with the first payment today. If the annual discount rate is 6%, the present value of the investment is closest to:

Correct answer: C

A. $72,098. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the present value of an annuity due with 9 payments instead of 10. Because the first payment occurs at t = 0, we can separate the annuity into two pieces: an immediate $10,000 that is received today (t = 0) and an ordinary annuity of $10,000 per year for 8 years. Using the formula for the present value of the ordinary annuity with 8 payments, we compute PV = A[1 – 1/(1 + r)N] / r = 10,000 × [1 – 1/(1+ 0.06)8]/0.06 = $62,097.94. Adding the two pieces, we obtain PV = $10,000 + $62,097.94 = $72,097.94 ≈ $72,098. Alternatively, using a financial calculator in BGN mode: N = 9; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = $72,097.94 ≈ $72,098.

Incorrect Answer Feedback: Incorrect because it is the present value of an annuity due with 9 payments instead of 10. Because the first payment occurs at $t = 0$, we can separate the annuity into two pieces: an immediate \$10,000 that is received today ($t = 0$) and an ordinary annuity of \$10,000 per year for 8 years.

Using the formula for the present value of the ordinary annuity with 8 payments, we compute $PV = A[1 - 1/(1 + r)^N] / r = 10{,}000 \times [1 - 1/(1+ 0.06)^8]/0.06 = \$62{,}097.94$.

Adding the two pieces, we obtain $PV = \$10{,}000 + \$62{,}097.94 = \$72{,}097.94 \approx \$72{,}098$.

Alternatively, using a financial calculator in BGN mode:

N = 9; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = \$72,097.94 ≈ \$72,098.

Figure 59: Source equation for question 85, option A

Figure 60: Source equation for question 85, option A

B. \$73,601. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the present value of a regular annuity with 10 payments, whereas the investment in the example is an annuity due. sing the formula for the present value of the ordinary annuity, we compute PV = A[1 – 1/(1 + r)N] / r = 10,000 × [1 – 1/(1+ 0.06)10]/0.06 = \$73,600.87 ≈ \$73,601. Alternative solution using a financial calculator in END mode: N = 10; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = \$73,600.87 ≈ \$73,601.

Incorrect Answer Feedback: Incorrect because this is the present value of a regular annuity with 10 payments, whereas the investment in the example is an annuity due. sing the formula for the present value of the ordinary annuity, we compute $PV = A[1 - 1/(1 + r)^N] / r = 10{,}000 \times [1 - 1/(1+ 0.06)^{10}]/0.06 = \$73{,}600.87 \approx \$73{,}601$.

Alternative solution using a financial calculator in END mode:

N = 10; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = \$73,600.87 ≈ \$73,601.

Figure 61: Source equation for question 85, option B

C. \$78,017. - Correct

Correct Answer Feedback: Correct because the investment represents an annuity due with 10 payments. Because the first payment occurs at t = 0, we can separate the annuity into two pieces: an immediate \$10,000 that is received today (t = 0) and an ordinary annuity of \$10,000 per year for 9 years. Using the formula for the present value of the ordinary annuity with 9 payments, we N 9 compute PV = A[1 – 1/(1 + r)] / r = \$10,000 × [1 – 1/(1+ 0.06)]/0.06 = >\$68,106.92. Adding the two pieces, we obtain PV = \$10,000 + \$68,106.92 = \$78,106.92 ≈ \$78,107. Alternatively, using a financial calculator with END mode: Compute the present value of the ordinary annuity with N = 9; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = \$68,016.92 Adding them, we obtain PV= \$10,000 + \$68,016.92 = \$78,016.92 ≈ \$78,017. Alternatively, using a financial calculator with BGN mode: N = 10; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = \$78,016.92 ≈ \$78,017.

Question 86

All else being equal, cash dividends paid to common shareholders result in a:

Correct answer: B

Correct Answer Feedback: Correct because the investment represents an annuity due with 10 payments. Because the first payment occurs at $t = 0$, we can separate the annuity into two pieces: an immediate $10,000 that is received today ($t = 0$) and an ordinary annuity of $10,000 per year for 9 years.

Using the formula for the present value of the ordinary annuity with 9 payments, we compute $PV = A[1 - 1/(1 + r)^N] / r = \$10,000 \times [1 - 1/(1+ 0.06)^9]/0.06 = >\$68,106.92$.

Adding the two pieces, we obtain $PV = \$10,000 + \$68,106.92 = \$78,106.92 \approx \$78,107$.

Alternatively, using a financial calculator with END mode:

Compute the present value of the ordinary annuity with N = 9; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = $68,016.92

Adding them, we obtain $PV = \$10,000 + \$68,016.92 = \$78,016.92 \approx \$78,017$.

Alternatively, using a financial calculator with BGN mode:

N = 10; I/Y = 6; PMT = 10,000; FV = 0; CPT PV = $78,016.92 ≈ $78,017.

(no answer)

Figure 62: Source equation for question 85, option C

A. lower ROA than if no dividends were paid. - Incorrect

Incorrect Answer Feedback: Incorrect because ROA = Net income / Average total assets. Declaring and paying cash dividends decreases total assets but does not affect net income. As a result, ROA increases compared to no dividends declared and paid.

Incorrect Answer Feedback: Incorrect because ROA = Net income / Average total assets. Declaring and paying cash dividends decreases total assets but does not affect net income. As a result, ROA increases compared to no dividends declared and paid.

Figure 63: Source equation for question 86, option A

Figure 64: Source equation for question 86, option A

B. higher ROE than if no dividends were paid. - Correct

Correct Answer Feedback: Correct because ROE = Net income / Average total equity. The basic components of owners' equity are paid- in capital and retained earnings. Retained earnings include the cumulative amount of the company's profits that have been retained in the company. For retained earnings, a dividend payment is the most common decrease. Declaring and paying cash dividends to shareholders does not change net income but reduces retained earnings thus decreases total equity. As a result, ROE will increase compared to no dividends declared and paid.

Correct Answer Feedback: Correct because ROE = Net income / Average total equity. The basic components of owners' equity are paid- in capital and retained earnings. Retained earnings include the cumulative amount of the company's profits that have been retained in the company. For retained earnings, a dividend payment is the most common decrease. Declaring and paying cash dividends to shareholders does not change net income but reduces retained earnings thus decreases total equity. As a result, ROE will increase compared to no dividends declared and paid.

Figure 65: Source equation for question 86, option B

C. lower net profit margin than if no dividends were paid. - Incorrect

Incorrect Answer Feedback: Incorrect because net profit margin = Net income / Revenue. Declaring and paying cash dividend affects neither net income nor revenue. As such, net profit margin remains unchanged.

Incorrect Answer Feedback: Incorrect because net profit margin = Net income / Revenue. Declaring and paying cash dividend affects neither net income nor revenue. As such, net profit margin remains unchanged.

(no answer)

Figure 66: Source equation for question 86, option C

Question 87

The following ten observations are a sample drawn from an approximately normal population:

Observation	1	2	3	4	5	6	7	8	9	10
Value	–31	–14	3	–18	34	20	–6	9	7	–16

The sample standard deviation is closest to:

Correct answer: C

A. 17.56. - Incorrect

Incorrect Answer Feedback: Incorrect. 17.56 is the sum of the absolute value of the deviations from mean divided by 9: Value Absolute deviation from mean –31 |–31 − (–1.2)| = 29.8 –14 |–14 − (–1.2)| = 12.8 3 |3 − (–1.2)| = 4.2 –18 |–18 − (–1.2)| = 16.8 34 |34 − (–1.2)| = 35.2 20 |20 − (–1.2)| = 21.2 –6 |–6 − (–1.2)| = 4.8 9 |9 − (–1.2)| = 10.2 7 |7 − (–1.2)| = 8.2 –16 |–16 − (–1.2)| = 14.8 Sum of absolute deviations 158 Divided by n – 1 (10 – 1) 158 / 9 = 17.56

Incorrect Answer Feedback: Incorrect. 17.56 is the sum of the absolute value of the deviations from mean divided by 9:

Value	Absolute deviation from mean
−31	|−31 − (−1.2)| = 29.8
−14	|−14 − (−1.2)| = 12.8
3	|3 − (−1.2)| = 4.2
−18	|−18 − (−1.2)| = 16.8
34	|34 − (−1.2)| = 35.2
20	|20 − (−1.2)| = 21.2
−6	|−6 − (−1.2)| = 4.8
9	|9 − (−1.2)| = 10.2
7	|7 − (−1.2)| = 8.2
−16	|−16 − (−1.2)| = 14.8
Sum of absolute deviations	158
Divided by n − 1 (10 − 1)	158 / 9 = 17.56

Figure 68: Source equation for question 87, option A

B. 18.58. - Incorrect

Incorrect Answer Feedback: Incorrect It is calculated by dividing the sum of squared deviations by 10 rather than 9: 3,453.60/10 = 345.36 and = 18.58.

Incorrect Answer Feedback: Incorrect It is calculated by dividing the sum of squared deviations by 10 rather than 9: 3,453.60/10 = 345.36 and $\sqrt{345.36}$ = **18.58.**

Figure 69: Source equation for question 87, option B

C. 19.59. - Correct

Correct Answer Feedback: Correct. = (−31 − 14 + 3 − 18 + 34 + 20 − 6 + 9 + 7 − 16)/10 = −12.00/10 = −1.20 where X = the value of the ith observation i n = the number of observations in the sample The sample variance is: The sample standard deviation is the (positive) square root of the sample variance: Value Deviation from Mean Squared Deviation –31 –31 − (–1.2) = –29.8 888.04 –14 –14 − (–1.2) = –12.8 163.84 3 3 − (–1.2) = 4.2 17.64 –18 –18 − (–1.2) = –16.8 282.24 34 34 − (–1.2) = 35.2 1,239.04 20 20 − (–1.2) = 21.2 449.44 –6 –6 − (–1.2) = –4.8 23.04 9 9 − (–1.2) = 10.2 104.04 7 7 − (–1.2) = 8.2 67.24 –16 –16 − (–1.2) = –14.8 219.04 Sum of squared deviations 3,453.60 Divided by n – 1 (10 – 1) 3,453.60 / 9 = 383.73 Square root √383.73 = 19.59

Question 88

The velocity of geopolitical risk is best described as the:

Correct answer: A

A. pace of its impact on the investor portfolio. - Correct

Correct Answer Feedback: Correct because the velocity of geopolitical risk is the pace at which it impacts an investor portfolio.

B. frequency of its impact on the investor portfolio. - Incorrect

Incorrect Answer Feedback: Incorrect because the velocity of geopolitical risk is the pace at which it impacts an investor portfolio, not the frequency of its impact.

C. magnitude of its impact on the investor portfolio. - Incorrect

Incorrect Answer Feedback: Incorrect because the velocity of geopolitical risk is the pace at which it impacts an investor portfolio, not the magnitude of its impact.

Question 89

If the USD/EUR (amount of USD per 1 EUR) spot rate is 1.2055 and the 12-month forward points are 21.3, the 12-month forward rate is closest to:

Correct answer: B

A. 1.20337. - Incorrect

Incorrect Answer Feedback: Incorrect because the number of points divided by 10,000 is subtracted from the spot exchange rate (instead of added): 1.2055 – 21.3/10,000 = 1.2055 – 0.00213 = 1.20337.

Correct Answer Feedback: Correct.

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

= (−31 − 14 + 3 − 18 + 34 + 20 − 6 + 9 + 7 − 16)/10

= −12.00/10 = −1.20

where

X_i = the value of the ith observation

n = the number of observations in the sample

The sample variance is:

$$s^2 = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{(n-1)}$$

The sample standard deviation is the (positive) square root of the sample variance:

Value	Deviation from Mean	Squared Deviation
−31	−31 − (−1.2) = −29.8	888.04
−14	−14 − (−1.2) = −12.8	163.84
3	3 − (−1.2) = 4.2	17.64
−18	−18 − (−1.2) = −16.8	282.24
34	34 − (−1.2) = 35.2	1,239.04
20	20 − (−1.2) = 21.2	449.44
−6	−6 − (−1.2) = −4.8	23.04
9	9 − (−1.2) = 10.2	104.04
7	7 − (−1.2) = 8.2	67.24
−16	−16 − (−1.2) = −14.8	219.04
Sum of squared deviations		3,453.60
Divided by n − 1 (10 − 1)		3,453.60 / 9 = 383.73
Square root		√383.73 = 19.59

(no answer)

Figure 70: Source equation for question 87, option C

Incorrect Answer Feedback: Incorrect because the number of points divided by 10,000 is subtracted from the spot exchange rate (instead of added): 1.2055 − 21.3/10,000 = 1.2055 − 0.00213 = 1.20337.

Figure 71: Source equation for question 89, option A

B. 1.20763. - Correct

Correct Answer Feedback: Correct because to convert any of these quoted forward points into a forward rate, one would divide the number of points by 10,000 (to scale down to the fourth decimal place, the last decimal place in the spot quote) and then add the result to the spot exchange rate quote. In this case: 1.2055 + (21.3/10,000) = 1.2055 + 0.00213 = 1.20763.

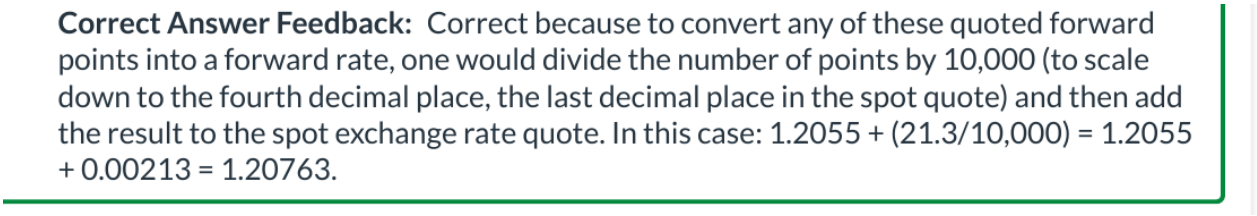


Figure 72: Source equation for question 89, option B

C. 1.22680. - Incorrect

Incorrect Answer Feedback: Incorrect because the number of points is divided by 1,000 instead of 10,000: 1.2055 + 21.3/1,000 = 1.2055 + 0.0213 = 1.22680

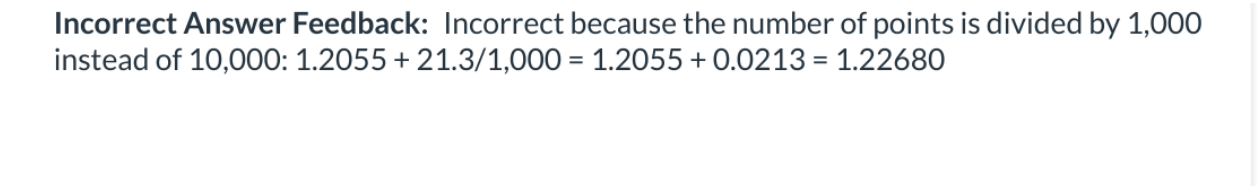


Figure 73: Source equation for question 89, option C



Figure 74: Source equation for question 89, option C

Question 90

An analyst gathers the following information about foreign exchange rates:

	Spot Rate
CAD/USD	1.2930
AUD/USD	1.3245

CAD/USD is the amount of CAD per 1 USD. AUD/USD is the amount of AUD per 1 USD. The spot CAD/AUD (amount of CAD per 1 AUD) cross-rate is closest to:

Correct answer: A

A. 0.9762. - Correct

Correct Answer Feedback: Correct because the CAD/AUD cross-rate = CAD/USD × (AUD/USD)-1 = 1.2930 / 1.3245 ≈ 0.9762.

B. 1.0244. - Incorrect

Incorrect Answer Feedback: Incorrect because the rates are inversely divided; AUD/USD / CAD/USD = 1.3245 / 1.2930 ≈ 1.0244.

Correct Answer Feedback: Correct because the CAD/AUD cross-rate = CAD/USD × $(AUD/USD)^{-1}$ = 1.2930 / 1.3245 ≈ 0.9762.

Figure 75: Source equation for question 90, option A

Incorrect Answer Feedback: Incorrect because the rates are inversely divided; AUD/USD / CAD/USD = 1.3245 / 1.2930 ≈ 1.0244.

Figure 76: Source equation for question 90, option B

C. 1.7126. - Incorrect

Incorrect Answer Feedback: Incorrect because the rates are incorrectly multiplied; CAD/USD × AUD/USD = 1.2930 × 1.3245 ≈ 1.7126.

Incorrect Answer Feedback: Incorrect because the rates are incorrectly multiplied; CAD/USD × AUD/USD = 1.2930 × 1.3245 ≈ 1.7126.

(no answer)

Figure 77: Source equation for question 90, option C